

Pinned group summary: SU-n6-q3-eps-minus1

Pinning-Sympy automated output

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1 Basic group attributes

Group	SU-n6-q3-eps-minus1
Matrix size	6×6
Rank	3
Defining equations	$X^*HX = H$ and $\det(X) = 1$

1.1 Skew-hermitian form

Dimension	6
Witt index	3
Primitive element	$\sqrt{d}$
Epsilon	-1

Matrix:
$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 \end{bmatrix}$$

1.2 Lie algebra

The defining equations for the Lie algebra are: $X^*H + HX = 0$ and $\text{tr}(X) = 0$

A generic element of the Lie algebra is:

$$\begin{bmatrix} \sqrt{d}x_{i00} + x_{r00} & \sqrt{d}x_{i01} + x_{r01} & \sqrt{d}x_{i02} + x_{r02} & -x_{r03} & -\sqrt{d}x_{i13} + x_{r13} & -\sqrt{d}x_{i23} + x_{r23} \\ \sqrt{d}x_{i10} + x_{r10} & \sqrt{d}x_{i11} + x_{r11} & \sqrt{d}x_{i12} + x_{r12} & \sqrt{d}x_{i13} + x_{r13} & -x_{r14} & -\sqrt{d}x_{i24} + x_{r24} \\ \sqrt{d}x_{i20} + x_{r20} & \sqrt{d}x_{i21} + x_{r21} & -\sqrt{d}(x_{i00} + x_{i11}) + x_{r22} & \sqrt{d}x_{i23} + x_{r23} & \sqrt{d}x_{i24} + x_{r24} & -x_{r25} \\ -x_{r30} & -\sqrt{d}x_{i40} + x_{r40} & -\sqrt{d}x_{i50} + x_{r50} & \sqrt{d}x_{i00} - x_{r00} & \sqrt{d}x_{i10} - x_{r10} & \sqrt{d}x_{i20} - x_{r20} \\ \sqrt{d}x_{i40} + x_{r40} & -x_{r41} & -\sqrt{d}x_{i51} + x_{r51} & \sqrt{d}x_{i01} - x_{r01} & \sqrt{d}x_{i11} - x_{r11} & \sqrt{d}x_{i21} - x_{r21} \\ \sqrt{d}x_{i50} + x_{r50} & \sqrt{d}x_{i51} + x_{r51} & -x_{r52} & \sqrt{d}x_{i02} - x_{r02} & \sqrt{d}x_{i12} - x_{r12} & -\sqrt{d}(x_{i00} + x_{i11}) - x_{r22} \end{bmatrix}$$

1.3 Torus

The torus has rank 3. A generic element of the torus is

$$\begin{bmatrix} t_0 & 0 & 0 & 0 & 0 & 0 \\ 0 & t_1 & 0 & 0 & 0 & 0 \\ 0 & 0 & t_2 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{t_0} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{t_1} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{t_2} \end{bmatrix}$$

The rows of the matrix below are trivial characters:

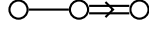
$$\begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

2 Root system

Dynkin type	C_3
Irreducible	True
Reduced	True
Simply laced	False
Number of roots	18

2.1 Dynkin diagram

The root system is irreducible and has Dynkin diagram:



2.2 Cartan matrix

The root system is irreducible and has Cartan matrix:

$$\begin{bmatrix} 2 & -2 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

2.3 Roots

Definition 2.1. A root $\alpha \in \Phi$ is **multipliable** if $2\alpha \in \Phi$.

Root	Squared norm	Simple	Sign	Multipliable
(0, 1, -1, 0, 0, 0)	2	Yes	+	No
(1, -1, 0, 0, 0, 0)	2	Yes	+	No
(0, 1, 1, 0, 0, 0)	2	No	+	No
(1, 0, -1, 0, 0, 0)	2	No	+	No
(1, 0, 1, 0, 0, 0)	2	No	+	No
(1, 1, 0, 0, 0, 0)	2	No	+	No
(-1, -1, 0, 0, 0, 0)	2	No	-	No
(-1, 0, -1, 0, 0, 0)	2	No	-	No
(-1, 0, 1, 0, 0, 0)	2	No	-	No
(-1, 1, 0, 0, 0, 0)	2	No	-	No
(0, -1, -1, 0, 0, 0)	2	No	-	No
(0, -1, 1, 0, 0, 0)	2	No	-	No
(0, 0, 2, 0, 0, 0)	4	Yes	+	No
(0, 2, 0, 0, 0, 0)	4	No	+	No
(2, 0, 0, 0, 0, 0)	4	No	+	No
(-2, 0, 0, 0, 0, 0)	4	No	-	No
(0, -2, 0, 0, 0, 0)	4	No	-	No
(0, 0, -2, 0, 0, 0)	4	No	-	No

2.4 Coroots

Root	Coroot
(-2, 0, 0, 0, 0, 0)	(-1, 0, 0, 1, 0, 0)
(-1, -1, 0, 0, 0, 0)	(-1, -1, 0, 1, 1, 0)
(-1, 0, -1, 0, 0, 0)	(-1, 0, -1, 1, 0, 1)
(-1, 0, 1, 0, 0, 0)	(-1, 0, 1, 1, 0, -1)
(-1, 1, 0, 0, 0, 0)	(-1, 1, 0, 1, -1, 0)
(0, -2, 0, 0, 0, 0)	(0, -1, 0, 0, 1, 0)
(0, -1, -1, 0, 0, 0)	(0, -1, -1, 0, 1, 1)
(0, -1, 1, 0, 0, 0)	(0, -1, 1, 0, 1, -1)
(0, 0, -2, 0, 0, 0)	(0, 0, -1, 0, 0, 1)
(0, 0, 2, 0, 0, 0)	(0, 0, 1, 0, 0, -1)
(0, 1, -1, 0, 0, 0)	(0, 1, -1, 0, -1, 1)
(0, 1, 1, 0, 0, 0)	(0, 1, 1, 0, -1, -1)
(0, 2, 0, 0, 0, 0)	(0, 1, 0, 0, -1, 0)
(1, -1, 0, 0, 0, 0)	(1, -1, 0, -1, 1, 0)
(1, 0, -1, 0, 0, 0)	(1, 0, -1, -1, 0, 1)
(1, 0, 1, 0, 0, 0)	(1, 0, 1, -1, 0, -1)
(1, 1, 0, 0, 0, 0)	(1, 1, 0, -1, -1, 0)
(2, 0, 0, 0, 0, 0)	(1, 0, 0, -1, 0, 0)

2.5 Linear combinations of roots

Definition 2.2. For two roots α, β in a root system Φ , the set of positive integral linear combinations of α, β that are also roots is denoted $(\alpha, \beta) = \{i\alpha + j\beta \in \Phi : i, j \in \mathbb{Z}_{\geq 1}\}$.

α	β	Pairs (i, j)
(-1, -1, 0, 0, 0, 0)	(-1, 1, 0, 0, 0, 0)	(1, 1)
(-1, -1, 0, 0, 0, 0)	(0, 1, -1, 0, 0, 0)	(1, 1)
(-1, -1, 0, 0, 0, 0)	(0, 1, 1, 0, 0, 0)	(1, 1)
(-1, -1, 0, 0, 0, 0)	(1, -1, 0, 0, 0, 0)	(1, 1)
(-1, -1, 0, 0, 0, 0)	(1, 0, -1, 0, 0, 0)	(1, 1)
(-1, -1, 0, 0, 0, 0)	(1, 0, 1, 0, 0, 0)	(1, 1)
(-1, 0, -1, 0, 0, 0)	(-1, 0, 1, 0, 0, 0)	(1, 1)
(-1, 0, -1, 0, 0, 0)	(0, -1, 1, 0, 0, 0)	(1, 1)
(-1, 0, -1, 0, 0, 0)	(0, 1, 1, 0, 0, 0)	(1, 1)
(-1, 0, -1, 0, 0, 0)	(1, -1, 0, 0, 0, 0)	(1, 1)
(-1, 0, -1, 0, 0, 0)	(1, 0, -1, 0, 0, 0)	(1, 1)
(-1, 0, -1, 0, 0, 0)	(1, 1, 0, 0, 0, 0)	(1, 1)
(-1, 0, 1, 0, 0, 0)	(0, -1, -1, 0, 0, 0)	(1, 1)
(-1, 0, 1, 0, 0, 0)	(0, 1, -1, 0, 0, 0)	(1, 1)
(-1, 0, 1, 0, 0, 0)	(1, -1, 0, 0, 0, 0)	(1, 1)
(-1, 0, 1, 0, 0, 0)	(1, 0, 1, 0, 0, 0)	(1, 1)
(-1, 0, 1, 0, 0, 0)	(1, 1, 0, 0, 0, 0)	(1, 1)
(-1, 1, 0, 0, 0, 0)	(0, -1, -1, 0, 0, 0)	(1, 1)
(-1, 1, 0, 0, 0, 0)	(0, -1, 1, 0, 0, 0)	(1, 1)
(-1, 1, 0, 0, 0, 0)	(1, 0, -1, 0, 0, 0)	(1, 1)
(-1, 1, 0, 0, 0, 0)	(1, 0, 1, 0, 0, 0)	(1, 1)
(-1, 1, 0, 0, 0, 0)	(1, 1, 0, 0, 0, 0)	(1, 1)
(0, -1, -1, 0, 0, 0)	(0, -1, 1, 0, 0, 0)	(1, 1)
(0, -1, -1, 0, 0, 0)	(0, 1, -1, 0, 0, 0)	(1, 1)

α	β	Pairs (i, j)
(0, -1, -1, 0, 0, 0)	(1, 0, 1, 0, 0, 0)	(1, 1)
(0, -1, -1, 0, 0, 0)	(1, 1, 0, 0, 0, 0)	(1, 1)
(0, -1, 1, 0, 0, 0)	(0, 1, 1, 0, 0, 0)	(1, 1)
(0, -1, 1, 0, 0, 0)	(1, 0, -1, 0, 0, 0)	(1, 1)
(0, -1, 1, 0, 0, 0)	(1, 1, 0, 0, 0, 0)	(1, 1)
(0, 1, -1, 0, 0, 0)	(0, 1, 1, 0, 0, 0)	(1, 1)
(0, 1, -1, 0, 0, 0)	(1, -1, 0, 0, 0, 0)	(1, 1)
(0, 1, -1, 0, 0, 0)	(1, 0, 1, 0, 0, 0)	(1, 1)
(0, 1, 1, 0, 0, 0)	(1, -1, 0, 0, 0, 0)	(1, 1)
(0, 1, 1, 0, 0, 0)	(1, 0, -1, 0, 0, 0)	(1, 1)
(1, -1, 0, 0, 0, 0)	(1, 1, 0, 0, 0, 0)	(1, 1)
(1, 0, -1, 0, 0, 0)	(1, 0, 1, 0, 0, 0)	(1, 1)
(-2, 0, 0, 0, 0, 0)	(1, -1, 0, 0, 0, 0)	(1, 1), (1, 2)
(-2, 0, 0, 0, 0, 0)	(1, 0, -1, 0, 0, 0)	(1, 1), (1, 2)
(-2, 0, 0, 0, 0, 0)	(1, 0, 1, 0, 0, 0)	(1, 1), (1, 2)
(-2, 0, 0, 0, 0, 0)	(1, 1, 0, 0, 0, 0)	(1, 1), (1, 2)
(-1, -1, 0, 0, 0, 0)	(0, 2, 0, 0, 0, 0)	(1, 1), (2, 1)
(-1, -1, 0, 0, 0, 0)	(2, 0, 0, 0, 0, 0)	(1, 1), (2, 1)
(-1, 0, -1, 0, 0, 0)	(0, 0, 2, 0, 0, 0)	(1, 1), (2, 1)
(-1, 0, -1, 0, 0, 0)	(2, 0, 0, 0, 0, 0)	(1, 1), (2, 1)
(-1, 0, 1, 0, 0, 0)	(0, 0, -2, 0, 0, 0)	(1, 1), (2, 1)
(-1, 0, 1, 0, 0, 0)	(2, 0, 0, 0, 0, 0)	(1, 1), (2, 1)
(-1, 1, 0, 0, 0, 0)	(0, -2, 0, 0, 0, 0)	(1, 1), (2, 1)
(-1, 1, 0, 0, 0, 0)	(2, 0, 0, 0, 0, 0)	(1, 1), (2, 1)
(0, -2, 0, 0, 0, 0)	(0, 1, -1, 0, 0, 0)	(1, 1), (1, 2)
(0, -2, 0, 0, 0, 0)	(0, 1, 1, 0, 0, 0)	(1, 1), (1, 2)
(0, -2, 0, 0, 0, 0)	(1, 1, 0, 0, 0, 0)	(1, 1), (1, 2)
(0, -1, -1, 0, 0, 0)	(0, 0, 2, 0, 0, 0)	(1, 1), (2, 1)
(0, -1, -1, 0, 0, 0)	(0, 2, 0, 0, 0, 0)	(1, 1), (2, 1)
(0, -1, 1, 0, 0, 0)	(0, 0, -2, 0, 0, 0)	(1, 1), (2, 1)
(0, -1, 1, 0, 0, 0)	(0, 2, 0, 0, 0, 0)	(1, 1), (2, 1)
(0, 0, -2, 0, 0, 0)	(0, 1, 1, 0, 0, 0)	(1, 1), (1, 2)
(0, 0, -2, 0, 0, 0)	(1, 0, 1, 0, 0, 0)	(1, 1), (1, 2)
(0, 0, 2, 0, 0, 0)	(0, 1, -1, 0, 0, 0)	(1, 1), (1, 2)
(0, 0, 2, 0, 0, 0)	(1, 0, -1, 0, 0, 0)	(1, 1), (1, 2)
(0, 2, 0, 0, 0, 0)	(1, -1, 0, 0, 0, 0)	(1, 1), (1, 2)

3 Root spaces

3.1 Dimensions

The table below lists each root along with the dimension of the associated root space (and root subgroup).

Root (α)	Dimension of root space (d_α)
$(-2, 0, 0, 0, 0, 0)$	1
$(0, -2, 0, 0, 0, 0)$	1
$(0, 0, -2, 0, 0, 0)$	1
$(0, 0, 2, 0, 0, 0)$	1
$(0, 2, 0, 0, 0, 0)$	1
$(2, 0, 0, 0, 0, 0)$	1
$(-1, -1, 0, 0, 0, 0)$	2
$(-1, 0, -1, 0, 0, 0)$	2
$(-1, 0, 1, 0, 0, 0)$	2
$(-1, 1, 0, 0, 0, 0)$	2
$(0, -1, -1, 0, 0, 0)$	2
$(0, -1, 1, 0, 0, 0)$	2
$(0, 1, -1, 0, 0, 0)$	2
$(0, 1, 1, 0, 0, 0)$	2
$(1, -1, 0, 0, 0, 0)$	2
$(1, 0, -1, 0, 0, 0)$	2
$(1, 0, 1, 0, 0, 0)$	2
$(1, 1, 0, 0, 0, 0)$	2

3.2 Root space table

The table below lists each root along with a generic element of the associated root space (inside the Lie algebra).
Include a definition here? INCOMPLETE

Root	Generic element of root space
$(-2, 0, 0, 0, 0, 0)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -u_0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
$(-1, -1, 0, 0, 0, 0)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\sqrt{d}u_1 + u_0 & 0 & 0 & 0 & 0 \\ \sqrt{d}u_1 + u_0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
$(-1, 0, -1, 0, 0, 0)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\sqrt{d}u_0 + u_1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ \sqrt{d}u_0 + u_1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

Root	Generic element of root space
$(-1, 0, 1, 0, 0, 0)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ \sqrt{d}u_1 + u_0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \sqrt{d}u_1 - u_0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
$(-1, 1, 0, 0, 0, 0)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ \sqrt{d}u_1 + u_0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \sqrt{d}u_1 - u_0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
$(0, -2, 0, 0, 0, 0)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -u_0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
$(0, -1, -1, 0, 0, 0)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\sqrt{d}u_0 + u_1 & 0 & 0 & 0 \\ 0 & \sqrt{d}u_0 + u_1 & 0 & 0 & 0 & 0 \end{bmatrix}$
$(0, -1, 1, 0, 0, 0)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \sqrt{d}u_0 + u_1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \sqrt{d}u_0 - u_1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
$(0, 0, -2, 0, 0, 0)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -u_0 & 0 & 0 & 0 \end{bmatrix}$
$(0, 0, 2, 0, 0, 0)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -u_0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
$(0, 1, -1, 0, 0, 0)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \sqrt{d}u_0 + u_1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \sqrt{d}u_0 - u_1 & 0 \end{bmatrix}$

Root	Generic element of root space
$(0, 1, 1, 0, 0, 0)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\sqrt{d}u_1 + u_0 \\ 0 & 0 & 0 & 0 & \sqrt{d}u_1 + u_0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
$(0, 2, 0, 0, 0, 0)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -u_0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
$(1, -1, 0, 0, 0, 0)$	$\begin{bmatrix} 0 & \sqrt{d}u_1 + u_0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sqrt{d}u_1 - u_0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
$(1, 0, -1, 0, 0, 0)$	$\begin{bmatrix} 0 & 0 & \sqrt{d}u_1 + u_0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sqrt{d}u_1 - u_0 & 0 & 0 \end{bmatrix}$
$(1, 0, 1, 0, 0, 0)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\sqrt{d}u_1 + u_0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sqrt{d}u_1 + u_0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
$(1, 1, 0, 0, 0, 0)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & -\sqrt{d}u_1 + u_0 & 0 \\ 0 & 0 & 0 & \sqrt{d}u_1 + u_0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
$(2, 0, 0, 0, 0, 0)$	$\begin{bmatrix} 0 & 0 & 0 & -u_0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

4 Root subgroups

4.1 Root subgroup table

The table below lists each root along with a generic element of the associated root subgroup.

Root	Generic element of root subgroup
$(-2, 0, 0, 0, 0, 0)$	$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ -u_0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$
$(-1, -1, 0, 0, 0, 0)$	$\begin{bmatrix} 1 & & & 0 & 0 & 0 & 0 \\ 0 & & & 1 & 0 & 0 & 0 \\ 0 & & & 0 & 1 & 0 & 0 \\ 0 & & -\sqrt{d}u_1 + u_0 & 0 & 1 & 0 & 0 \\ \sqrt{d}u_1 + u_0 & & 0 & 0 & 0 & 1 & 0 \\ 0 & & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$
$(-1, 0, -1, 0, 0, 0)$	$\begin{bmatrix} 1 & 0 & & 0 & 0 & 0 & 0 \\ 0 & 1 & & 0 & 0 & 0 & 0 \\ 0 & 0 & & 1 & 0 & 0 & 0 \\ 0 & 0 & -\sqrt{d}u_0 + u_1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ \sqrt{d}u_0 + u_1 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$
$(-1, 0, 1, 0, 0, 0)$	$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ \sqrt{d}u_1 + u_0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & \sqrt{d}u_1 - u_0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$
$(-1, 1, 0, 0, 0, 0)$	$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ \sqrt{d}u_1 + u_0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & \sqrt{d}u_1 - u_0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$
$(0, -2, 0, 0, 0, 0)$	$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & -u_0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$
$(0, -1, -1, 0, 0, 0)$	$\begin{bmatrix} 1 & 0 & & 0 & 0 & 0 & 0 \\ 0 & 1 & & 0 & 0 & 0 & 0 \\ 0 & 0 & & 1 & 0 & 0 & 0 \\ 0 & 0 & & 0 & 1 & 0 & 0 \\ 0 & 0 & -\sqrt{d}u_0 + u_1 & 0 & 1 & 0 & 0 \\ 0 & \sqrt{d}u_0 + u_1 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$

Root	Generic element of root subgroup
$(0, -1, 1, 0, 0, 0)$	$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & \sqrt{d}u_0 + u_1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & \sqrt{d}u_0 - u_1 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$
$(0, 0, -2, 0, 0, 0)$	$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & -u_0 & 0 & 0 & 1 \end{bmatrix}$
$(0, 0, 2, 0, 0, 0)$	$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & -u_0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$
$(0, 1, -1, 0, 0, 0)$	$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & \sqrt{d}u_0 + u_1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & \sqrt{d}u_0 - u_1 & 1 \end{bmatrix}$
$(0, 1, 1, 0, 0, 0)$	$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & -\sqrt{d}u_1 + u_0 \\ 0 & 0 & 1 & 0 & \sqrt{d}u_1 + u_0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$
$(0, 2, 0, 0, 0, 0)$	$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -u_0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$
$(1, -1, 0, 0, 0, 0)$	$\begin{bmatrix} 1 & \sqrt{d}u_1 + u_0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & \sqrt{d}u_1 - u_0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$
$(1, 0, -1, 0, 0, 0)$	$\begin{bmatrix} 1 & 0 & \sqrt{d}u_1 + u_0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & \sqrt{d}u_1 - u_0 & 0 & 1 \end{bmatrix}$

Root	Generic element of root subgroup
$(1, 0, 1, 0, 0, 0)$	$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & -\sqrt{d}u_1 + u_0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & \sqrt{d}u_1 + u_0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$
$(1, 1, 0, 0, 0, 0)$	$\begin{bmatrix} 1 & 0 & 0 & 0 & -\sqrt{d}u_1 + u_0 & 0 \\ 0 & 1 & 0 & \sqrt{d}u_1 + u_0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$
$(2, 0, 0, 0, 0, 0)$	$\begin{bmatrix} 1 & 0 & 0 & -u_0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$

4.2 Homomorphism property

For each root $\alpha \in \Phi$, the associated root subgroup map $X_\alpha : V_\alpha \rightarrow G$ is a pseudo-homomorphism with respect to the additive structure on V_α and the group operation (i.e. matrix multiplication) in G . That is, for any $u, v \in V_\alpha$,

$$X_\alpha(u) \cdot X_\alpha(v) = X_\alpha(u + v) \cdot \prod_{\substack{i > 1 \\ i\alpha \in \Phi}} X_{i\alpha}(q_\alpha^i(u, v)) \quad (1)$$

for some homogeneous polynomial maps $q_\alpha^i : V_\alpha \times V_\alpha \rightarrow V_{i\alpha}$. Thus X_α is a group homomorphism if and only the product on the right hand side reduces to 1. The quantities $q_\alpha^i(u, v)$ are called **homomorphism defect coefficients**. For any root α of a root system Φ of any group considered in the Pinning-Sympy package, the set

$$\{i\alpha \in \Phi : i \in \mathbb{Z}_{\geq 2}\}$$

is either empty or a singleton set consisting of $\{2\alpha\}$. The non-empty case only arises when Φ is a non-reduced root system (of Dynkin type BC), and when α is a multipliable root.

Since the root system for this group is reduced, there are no multipliable roots and the product is always empty, hence each X_α is a group homomorphism there are no homomorphism defect coefficients.

5 Commutators

The Steinberg/Chevalley commutator formula says that for two roots $\alpha, \beta \in \Phi$, the commutator of two root group elements $X_\alpha(u)$ and $X_\beta(v)$ is controlled by the structure of Φ , in particular by which positive integer linear combinations $i\alpha + j\beta$ are elements of Φ . Recall that

$$(\alpha, \beta) = \{i\alpha + j\beta \in \Phi : i, j \in \mathbb{Z}_{\geq 1}\}$$

Specifically, we have the commutator formula

$$\left[X_\alpha(u), X_\beta(v)\right] = \prod_{(\alpha, \beta)} X_{i\alpha + j\beta} \left(N_{ij}^{\alpha\beta}(u, v)\right)$$

for some homogeneous polynomial functions $N_{ij}^{\alpha\beta} : V_\alpha \times V_\beta \rightarrow V_{i\alpha + j\beta}$. We repeat here the table of linear combinations of roots. The quantity $N_{ij}^{\alpha\beta}(u, v)$ is called a **commutator coefficient**. The table below gives these coefficients.

α	β	i	j	$i\alpha + j\beta$	$N_{ij}^{\alpha\beta}(u, v)$
$(-1, -1, 0, 0, 0, 0)$	$(-1, 1, 0, 0, 0, 0)$	1	1	$(-2, 0, 0, 0, 0, 0)$	$2du_1v_1 - 2u_0v_0$
$(-1, -1, 0, 0, 0, 0)$	$(0, 1, -1, 0, 0, 0)$	1	1	$(-1, 0, -1, 0, 0, 0)$	$\begin{bmatrix} -du_1v_1 + u_0v_0 & -u_0v_1 + u_1v_0 \end{bmatrix}$
$(-1, -1, 0, 0, 0, 0)$	$(0, 1, 1, 0, 0, 0)$	1	1	$(-1, 0, 1, 0, 0, 0)$	$\begin{bmatrix} -du_1v_0 - u_0v_1 & -u_0v_0 - u_1v_1 \end{bmatrix}$
$(-1, -1, 0, 0, 0, 0)$	$(1, -1, 0, 0, 0, 0)$	1	1	$(0, -2, 0, 0, 0, 0)$	$-2du_1v_0 - 2u_0v_1$
$(-1, -1, 0, 0, 0, 0)$	$(1, 0, -1, 0, 0, 0)$	1	1	$(0, -1, -1, 0, 0, 0)$	$\begin{bmatrix} -u_0v_0 - u_1v_1 & du_1v_0 + u_0v_1 \end{bmatrix}$
$(-1, -1, 0, 0, 0, 0)$	$(1, 0, 1, 0, 0, 0)$	1	1	$(0, -1, 1, 0, 0, 0)$	$\begin{bmatrix} du_1v_1 - u_0v_0 & -u_0v_1 + u_1v_0 \end{bmatrix}$
$(-1, 0, -1, 0, 0, 0)$	$(-1, 0, 1, 0, 0, 0)$	1	1	$(-2, 0, 0, 0, 0, 0)$	$2du_1v_1 - 2u_0v_0$
$(-1, 0, -1, 0, 0, 0)$	$(0, -1, 1, 0, 0, 0)$	1	1	$(-1, -1, 0, 0, 0, 0)$	$\begin{bmatrix} -du_1v_1 + u_0v_0 & -u_0v_1 + u_1v_0 \end{bmatrix}$
$(-1, 0, -1, 0, 0, 0)$	$(0, 1, 1, 0, 0, 0)$	1	1	$(-1, 1, 0, 0, 0, 0)$	$\begin{bmatrix} du_1v_0 - u_0v_1 & u_0v_0 - u_1v_1 \end{bmatrix}$
$(-1, 0, -1, 0, 0, 0)$	$(1, -1, 0, 0, 0, 0)$	1	1	$(0, -1, -1, 0, 0, 0)$	$\begin{bmatrix} u_0v_0 + u_1v_1 & du_1v_0 + u_0v_1 \end{bmatrix}$
$(-1, 0, -1, 0, 0, 0)$	$(1, 0, -1, 0, 0, 0)$	1	1	$(0, 0, -2, 0, 0, 0)$	$-2du_1v_0 - 2u_0v_1$
$(-1, 0, -1, 0, 0, 0)$	$(1, 1, 0, 0, 0, 0)$	1	1	$(0, 1, -1, 0, 0, 0)$	$\begin{bmatrix} du_1v_1 - u_0v_0 & -u_0v_1 + u_1v_0 \end{bmatrix}$
$(-1, 0, 1, 0, 0, 0)$	$(-1, 0, -1, 0, 0, 0)$	1	1	$(-2, 0, 0, 0, 0, 0)$	$-2du_1v_1 + 2u_0v_0$
$(-1, 0, 1, 0, 0, 0)$	$(0, -1, -1, 0, 0, 0)$	1	1	$(-1, -1, 0, 0, 0, 0)$	$\begin{bmatrix} du_1v_0 - u_0v_1 & u_0v_0 - u_1v_1 \end{bmatrix}$
$(-1, 0, 1, 0, 0, 0)$	$(0, 1, -1, 0, 0, 0)$	1	1	$(-1, 1, 0, 0, 0, 0)$	$\begin{bmatrix} -du_1v_1 - u_0v_0 & -u_0v_1 - u_1v_0 \end{bmatrix}$
$(-1, 0, 1, 0, 0, 0)$	$(1, -1, 0, 0, 0, 0)$	1	1	$(0, -1, 1, 0, 0, 0)$	$\begin{bmatrix} du_1v_0 + u_0v_1 & u_0v_0 + u_1v_1 \end{bmatrix}$
$(-1, 0, 1, 0, 0, 0)$	$(1, 0, 1, 0, 0, 0)$	1	1	$(0, 0, 2, 0, 0, 0)$	$2du_1v_1 - 2u_0v_0$
$(-1, 0, 1, 0, 0, 0)$	$(1, 1, 0, 0, 0, 0)$	1	1	$(0, 1, 1, 0, 0, 0)$	$\begin{bmatrix} -u_0v_1 + u_1v_0 & -du_1v_1 + u_0v_0 \end{bmatrix}$
$(-1, 1, 0, 0, 0, 0)$	$(-1, -1, 0, 0, 0, 0)$	1	1	$(-2, 0, 0, 0, 0, 0)$	$-2du_1v_1 + 2u_0v_0$
$(-1, 1, 0, 0, 0, 0)$	$(0, -1, -1, 0, 0, 0)$	1	1	$(-1, 0, -1, 0, 0, 0)$	$\begin{bmatrix} -du_1v_0 - u_0v_1 & -u_0v_0 - u_1v_1 \end{bmatrix}$
$(-1, 1, 0, 0, 0, 0)$	$(0, -1, 1, 0, 0, 0)$	1	1	$(-1, 0, 1, 0, 0, 0)$	$\begin{bmatrix} -du_1v_1 - u_0v_0 & -u_0v_1 - u_1v_0 \end{bmatrix}$
$(-1, 1, 0, 0, 0, 0)$	$(1, 0, -1, 0, 0, 0)$	1	1	$(0, 1, -1, 0, 0, 0)$	$\begin{bmatrix} du_1v_0 + u_0v_1 & u_0v_0 + u_1v_1 \end{bmatrix}$
$(-1, 1, 0, 0, 0, 0)$	$(1, 0, 1, 0, 0, 0)$	1	1	$(0, 1, 1, 0, 0, 0)$	$\begin{bmatrix} u_0v_1 - u_1v_0 & -du_1v_1 + u_0v_0 \end{bmatrix}$
$(-1, 1, 0, 0, 0, 0)$	$(1, 1, 0, 0, 0, 0)$	1	1	$(0, 2, 0, 0, 0, 0)$	$2du_1v_1 - 2u_0v_0$
$(0, -1, -1, 0, 0, 0)$	$(-1, 0, 1, 0, 0, 0)$	1	1	$(-1, -1, 0, 0, 0, 0)$	$\begin{bmatrix} -du_0v_1 + u_1v_0 & -u_0v_0 + u_1v_1 \end{bmatrix}$
$(0, -1, -1, 0, 0, 0)$	$(-1, 1, 0, 0, 0, 0)$	1	1	$(-1, 0, -1, 0, 0, 0)$	$\begin{bmatrix} du_0v_1 + u_1v_0 & u_0v_0 + u_1v_1 \end{bmatrix}$
$(0, -1, -1, 0, 0, 0)$	$(0, -1, 1, 0, 0, 0)$	1	1	$(0, -2, 0, 0, 0, 0)$	$2du_0v_1 - 2u_1v_0$
$(0, -1, -1, 0, 0, 0)$	$(0, 1, -1, 0, 0, 0)$	1	1	$(0, 0, -2, 0, 0, 0)$	$-2du_0v_1 - 2u_1v_0$
$(0, -1, -1, 0, 0, 0)$	$(1, 0, 1, 0, 0, 0)$	1	1	$(1, -1, 0, 0, 0, 0)$	$\begin{bmatrix} -u_0v_0 + u_1v_1 & du_0v_1 - u_1v_0 \end{bmatrix}$
$(0, -1, -1, 0, 0, 0)$	$(1, 1, 0, 0, 0, 0)$	1	1	$(1, 0, -1, 0, 0, 0)$	$\begin{bmatrix} u_0v_0 + u_1v_1 & -du_0v_1 - u_1v_0 \end{bmatrix}$
$(0, -1, 1, 0, 0, 0)$	$(-1, 0, -1, 0, 0, 0)$	1	1	$(-1, -1, 0, 0, 0, 0)$	$\begin{bmatrix} du_1v_1 - u_0v_0 & -u_0v_1 + u_1v_0 \end{bmatrix}$
$(0, -1, 1, 0, 0, 0)$	$(-1, 1, 0, 0, 0, 0)$	1	1	$(-1, 0, 1, 0, 0, 0)$	$\begin{bmatrix} du_1v_1 + u_0v_0 & u_0v_1 + u_1v_0 \end{bmatrix}$
$(0, -1, 1, 0, 0, 0)$	$(0, -1, -1, 0, 0, 0)$	1	1	$(0, -2, 0, 0, 0, 0)$	$-2du_1v_0 + 2u_0v_1$
$(0, -1, 1, 0, 0, 0)$	$(0, 1, 1, 0, 0, 0)$	1	1	$(0, 0, 2, 0, 0, 0)$	$2du_1v_0 - 2u_0v_1$

α	β	i	j	$i\alpha + j\beta$	$N_{ij}^{\alpha\beta}(u, v)$
(0, -1, 1, 0, 0, 0)	(1, 0, -1, 0, 0, 0)	1	1	(1, -1, 0, 0, 0, 0)	$[-u_0v_0 - u_1v_1 \quad -du_1v_0 - u_0v_1]$
(0, -1, 1, 0, 0, 0)	(1, 1, 0, 0, 0, 0)	1	1	(1, 0, 1, 0, 0, 0)	$[du_1v_1 + u_0v_0 \quad u_0v_1 + u_1v_0]$
(0, 1, -1, 0, 0, 0)	(-1, -1, 0, 0, 0, 0)	1	1	(-1, 0, -1, 0, 0, 0)	$[du_1v_1 - u_0v_0 \quad -u_0v_1 + u_1v_0]$
(0, 1, -1, 0, 0, 0)	(-1, 0, 1, 0, 0, 0)	1	1	(-1, 1, 0, 0, 0, 0)	$[du_1v_1 + u_0v_0 \quad u_0v_1 + u_1v_0]$
(0, 1, -1, 0, 0, 0)	(0, -1, -1, 0, 0, 0)	1	1	(0, 0, -2, 0, 0, 0)	$2du_1v_0 + 2u_0v_1$
(0, 1, -1, 0, 0, 0)	(0, 1, 1, 0, 0, 0)	1	1	(0, 2, 0, 0, 0, 0)	$-2du_1v_0 - 2u_0v_1$
(0, 1, -1, 0, 0, 0)	(1, -1, 0, 0, 0, 0)	1	1	(1, 0, -1, 0, 0, 0)	$[-u_0v_0 - u_1v_1 \quad -du_1v_0 - u_0v_1]$
(0, 1, -1, 0, 0, 0)	(1, 0, 1, 0, 0, 0)	1	1	(1, 1, 0, 0, 0, 0)	$[du_1v_1 + u_0v_0 \quad u_0v_1 + u_1v_0]$
(0, 1, 1, 0, 0, 0)	(-1, -1, 0, 0, 0, 0)	1	1	(-1, 0, 1, 0, 0, 0)	$[du_0v_1 + u_1v_0 \quad u_0v_0 + u_1v_1]$
(0, 1, 1, 0, 0, 0)	(-1, 0, -1, 0, 0, 0)	1	1	(-1, 1, 0, 0, 0, 0)	$[-du_0v_1 + u_1v_0 \quad -u_0v_0 + u_1v_1]$
(0, 1, 1, 0, 0, 0)	(0, -1, 1, 0, 0, 0)	1	1	(0, 0, 2, 0, 0, 0)	$-2du_0v_1 + 2u_1v_0$
(0, 1, 1, 0, 0, 0)	(0, 1, -1, 0, 0, 0)	1	1	(0, 2, 0, 0, 0, 0)	$2du_0v_1 + 2u_1v_0$
(0, 1, 1, 0, 0, 0)	(1, -1, 0, 0, 0, 0)	1	1	(1, 0, 1, 0, 0, 0)	$[du_0v_0 - u_1v_1 \quad -u_0v_1 + u_1v_0]$
(0, 1, 1, 0, 0, 0)	(1, 0, -1, 0, 0, 0)	1	1	(1, 1, 0, 0, 0, 0)	$[-du_0v_0 - u_1v_1 \quad u_0v_1 + u_1v_0]$
(1, -1, 0, 0, 0, 0)	(-1, -1, 0, 0, 0, 0)	1	1	(0, -2, 0, 0, 0, 0)	$2du_0v_1 + 2u_1v_0$
(1, -1, 0, 0, 0, 0)	(-1, 0, -1, 0, 0, 0)	1	1	(0, -1, -1, 0, 0, 0)	$[-u_0v_0 - u_1v_1 \quad -du_0v_1 - u_1v_0]$
(1, -1, 0, 0, 0, 0)	(-1, 0, 1, 0, 0, 0)	1	1	(0, -1, 1, 0, 0, 0)	$[-du_0v_1 - u_1v_0 \quad -u_0v_0 - u_1v_1]$
(1, -1, 0, 0, 0, 0)	(0, 1, -1, 0, 0, 0)	1	1	(1, 0, -1, 0, 0, 0)	$[u_0v_0 + u_1v_1 \quad du_0v_1 + u_1v_0]$
(1, -1, 0, 0, 0, 0)	(0, 1, 1, 0, 0, 0)	1	1	(1, 0, 1, 0, 0, 0)	$[-du_0v_0 + u_1v_1 \quad -u_0v_1 + u_1v_0]$
(1, -1, 0, 0, 0, 0)	(1, 1, 0, 0, 0, 0)	1	1	(2, 0, 0, 0, 0, 0)	$-2du_0v_1 - 2u_1v_0$
(1, 0, -1, 0, 0, 0)	(-1, -1, 0, 0, 0, 0)	1	1	(0, -1, -1, 0, 0, 0)	$[u_0v_0 + u_1v_1 \quad -du_0v_1 - u_1v_0]$
(1, 0, -1, 0, 0, 0)	(-1, 0, -1, 0, 0, 0)	1	1	(0, 0, -2, 0, 0, 0)	$2du_0v_1 + 2u_1v_0$
(1, 0, -1, 0, 0, 0)	(-1, 1, 0, 0, 0, 0)	1	1	(0, 1, -1, 0, 0, 0)	$[-du_0v_1 - u_1v_0 \quad -u_0v_0 - u_1v_1]$
(1, 0, -1, 0, 0, 0)	(0, -1, 1, 0, 0, 0)	1	1	(1, -1, 0, 0, 0, 0)	$[u_0v_0 + u_1v_1 \quad du_0v_1 + u_1v_0]$
(1, 0, -1, 0, 0, 0)	(0, 1, 1, 0, 0, 0)	1	1	(1, 1, 0, 0, 0, 0)	$[du_0v_0 + u_1v_1 \quad -u_0v_1 - u_1v_0]$
(1, 0, -1, 0, 0, 0)	(1, 0, 1, 0, 0, 0)	1	1	(2, 0, 0, 0, 0, 0)	$-2du_0v_1 - 2u_1v_0$
(1, 0, 1, 0, 0, 0)	(-1, -1, 0, 0, 0, 0)	1	1	(0, -1, 1, 0, 0, 0)	$[-du_1v_1 + u_0v_0 \quad -u_0v_1 + u_1v_0]$
(1, 0, 1, 0, 0, 0)	(-1, 0, 1, 0, 0, 0)	1	1	(0, 0, 2, 0, 0, 0)	$-2du_1v_1 + 2u_0v_0$
(1, 0, 1, 0, 0, 0)	(-1, 1, 0, 0, 0, 0)	1	1	(0, 1, 1, 0, 0, 0)	$[u_0v_1 - u_1v_0 \quad du_1v_1 - u_0v_0]$
(1, 0, 1, 0, 0, 0)	(0, -1, -1, 0, 0, 0)	1	1	(1, -1, 0, 0, 0, 0)	$[u_0v_0 - u_1v_1 \quad -du_1v_0 + u_0v_1]$
(1, 0, 1, 0, 0, 0)	(0, 1, -1, 0, 0, 0)	1	1	(1, 1, 0, 0, 0, 0)	$[-du_1v_1 - u_0v_0 \quad -u_0v_1 - u_1v_0]$
(1, 0, 1, 0, 0, 0)	(1, 0, -1, 0, 0, 0)	1	1	(2, 0, 0, 0, 0, 0)	$2du_1v_0 + 2u_0v_1$
(1, 1, 0, 0, 0, 0)	(-1, 0, -1, 0, 0, 0)	1	1	(0, 1, -1, 0, 0, 0)	$[-du_1v_1 + u_0v_0 \quad -u_0v_1 + u_1v_0]$
(1, 1, 0, 0, 0, 0)	(-1, 0, 1, 0, 0, 0)	1	1	(0, 1, 1, 0, 0, 0)	$[-u_0v_1 + u_1v_0 \quad du_1v_1 - u_0v_0]$
(1, 1, 0, 0, 0, 0)	(-1, 1, 0, 0, 0, 0)	1	1	(0, 2, 0, 0, 0, 0)	$-2du_1v_1 + 2u_0v_0$
(1, 1, 0, 0, 0, 0)	(0, -1, -1, 0, 0, 0)	1	1	(1, 0, -1, 0, 0, 0)	$[-u_0v_0 - u_1v_1 \quad du_1v_0 + u_0v_1]$
(1, 1, 0, 0, 0, 0)	(0, -1, 1, 0, 0, 0)	1	1	(1, 0, 1, 0, 0, 0)	$[-du_1v_1 - u_0v_0 \quad -u_0v_1 - u_1v_0]$
(1, 1, 0, 0, 0, 0)	(1, -1, 0, 0, 0, 0)	1	1	(2, 0, 0, 0, 0, 0)	$2du_1v_0 + 2u_0v_1$
(-2, 0, 0, 0, 0, 0)	(1, -1, 0, 0, 0, 0)	1	1	(-1, -1, 0, 0, 0, 0)	$[-u_0v_1 \quad u_0v_0]$
(-2, 0, 0, 0, 0, 0)	(1, -1, 0, 0, 0, 0)	1	2	(0, -2, 0, 0, 0, 0)	$u_0 (dv_0^2 - v_1^2)$
(-2, 0, 0, 0, 0, 0)	(1, 0, -1, 0, 0, 0)	1	1	(-1, 0, -1, 0, 0, 0)	$[-u_0v_1 \quad u_0v_0]$
(-2, 0, 0, 0, 0, 0)	(1, 0, -1, 0, 0, 0)	1	2	(0, 0, -2, 0, 0, 0)	$u_0 (dv_0^2 - v_1^2)$
(-2, 0, 0, 0, 0, 0)	(1, 0, 1, 0, 0, 0)	1	1	(-1, 0, 1, 0, 0, 0)	$[u_0v_0 \quad u_0v_1]$
(-2, 0, 0, 0, 0, 0)	(1, 0, 1, 0, 0, 0)	1	2	(0, 0, 2, 0, 0, 0)	$u_0 (-dv_1^2 + v_0^2)$
(-2, 0, 0, 0, 0, 0)	(1, 1, 0, 0, 0, 0)	1	1	(-1, 1, 0, 0, 0, 0)	$[u_0v_0 \quad u_0v_1]$
(-2, 0, 0, 0, 0, 0)	(1, 1, 0, 0, 0, 0)	1	2	(0, 2, 0, 0, 0, 0)	$u_0 (-dv_1^2 + v_0^2)$
(-1, -1, 0, 0, 0, 0)	(0, 2, 0, 0, 0, 0)	1	1	(-1, 1, 0, 0, 0, 0)	$[u_0v_0 \quad u_1v_0]$
(-1, -1, 0, 0, 0, 0)	(0, 2, 0, 0, 0, 0)	2	1	(-2, 0, 0, 0, 0, 0)	$v_0 (du_1^2 - u_0^2)$
(-1, -1, 0, 0, 0, 0)	(2, 0, 0, 0, 0, 0)	1	1	(1, -1, 0, 0, 0, 0)	$[-u_1v_0 \quad u_0v_0]$
(-1, -1, 0, 0, 0, 0)	(2, 0, 0, 0, 0, 0)	2	1	(0, -2, 0, 0, 0, 0)	$v_0 (du_1^2 - u_0^2)$

α	β	i	j	$i\alpha + j\beta$	$N_{ij}^{\alpha\beta}(u, v)$
$(-1, 0, -1, 0, 0, 0)$	$(0, 0, 2, 0, 0, 0)$	1	1	$(-1, 0, 1, 0, 0, 0)$	$[u_0v_0 \quad u_1v_0]$
$(-1, 0, -1, 0, 0, 0)$	$(0, 0, 2, 0, 0, 0)$	2	1	$(-2, 0, 0, 0, 0, 0)$	$v_0(du_1^2 - u_0^2)$
$(-1, 0, -1, 0, 0, 0)$	$(2, 0, 0, 0, 0, 0)$	1	1	$(1, 0, -1, 0, 0, 0)$	$[-u_1v_0 \quad u_0v_0]$
$(-1, 0, -1, 0, 0, 0)$	$(2, 0, 0, 0, 0, 0)$	2	1	$(0, 0, -2, 0, 0, 0)$	$v_0(du_1^2 - u_0^2)$
$(-1, 0, 1, 0, 0, 0)$	$(0, 0, -2, 0, 0, 0)$	1	1	$(-1, 0, -1, 0, 0, 0)$	$[u_0v_0 \quad u_1v_0]$
$(-1, 0, 1, 0, 0, 0)$	$(0, 0, -2, 0, 0, 0)$	2	1	$(-2, 0, 0, 0, 0, 0)$	$v_0(-du_1^2 + u_0^2)$
$(-1, 0, 1, 0, 0, 0)$	$(2, 0, 0, 0, 0, 0)$	1	1	$(1, 0, 1, 0, 0, 0)$	$[-u_0v_0 \quad -u_1v_0]$
$(-1, 0, 1, 0, 0, 0)$	$(2, 0, 0, 0, 0, 0)$	2	1	$(0, 0, 2, 0, 0, 0)$	$v_0(-du_1^2 + u_0^2)$
$(-1, 1, 0, 0, 0, 0)$	$(0, -2, 0, 0, 0, 0)$	1	1	$(-1, -1, 0, 0, 0, 0)$	$[u_0v_0 \quad u_1v_0]$
$(-1, 1, 0, 0, 0, 0)$	$(0, -2, 0, 0, 0, 0)$	2	1	$(-2, 0, 0, 0, 0, 0)$	$v_0(-du_1^2 + u_0^2)$
$(-1, 1, 0, 0, 0, 0)$	$(2, 0, 0, 0, 0, 0)$	1	1	$(1, 1, 0, 0, 0, 0)$	$[-u_0v_0 \quad -u_1v_0]$
$(-1, 1, 0, 0, 0, 0)$	$(2, 0, 0, 0, 0, 0)$	2	1	$(0, 2, 0, 0, 0, 0)$	$v_0(-du_1^2 + u_0^2)$
$(0, -2, 0, 0, 0, 0)$	$(-1, 1, 0, 0, 0, 0)$	1	1	$(-1, -1, 0, 0, 0, 0)$	$[-u_0v_0 \quad -u_0v_1]$
$(0, -2, 0, 0, 0, 0)$	$(-1, 1, 0, 0, 0, 0)$	1	2	$(-2, 0, 0, 0, 0, 0)$	$u_0(dv_1^2 - v_0^2)$
$(0, -2, 0, 0, 0, 0)$	$(0, 1, -1, 0, 0, 0)$	1	1	$(0, -1, -1, 0, 0, 0)$	$[u_0v_1 \quad -u_0v_0]$
$(0, -2, 0, 0, 0, 0)$	$(0, 1, -1, 0, 0, 0)$	1	2	$(0, 0, -2, 0, 0, 0)$	$u_0(dv_1^2 - v_0^2)$
$(0, -2, 0, 0, 0, 0)$	$(0, 1, 1, 0, 0, 0)$	1	1	$(0, -1, 1, 0, 0, 0)$	$[u_0v_1 \quad u_0v_0]$
$(0, -2, 0, 0, 0, 0)$	$(0, 1, 1, 0, 0, 0)$	1	2	$(0, 0, 2, 0, 0, 0)$	$u_0(-dv_0^2 + v_1^2)$
$(0, -2, 0, 0, 0, 0)$	$(1, 1, 0, 0, 0, 0)$	1	1	$(1, -1, 0, 0, 0, 0)$	$[-u_0v_1 \quad u_0v_0]$
$(0, -2, 0, 0, 0, 0)$	$(1, 1, 0, 0, 0, 0)$	1	2	$(2, 0, 0, 0, 0, 0)$	$u_0(-dv_1^2 + v_0^2)$
$(0, -1, -1, 0, 0, 0)$	$(0, 0, 2, 0, 0, 0)$	1	1	$(0, -1, 1, 0, 0, 0)$	$[u_1v_0 \quad u_0v_0]$
$(0, -1, -1, 0, 0, 0)$	$(0, 0, 2, 0, 0, 0)$	2	1	$(0, -2, 0, 0, 0, 0)$	$v_0(du_0^2 - u_1^2)$
$(0, -1, -1, 0, 0, 0)$	$(0, 2, 0, 0, 0, 0)$	1	1	$(0, 1, -1, 0, 0, 0)$	$[u_1v_0 \quad -u_0v_0]$
$(0, -1, -1, 0, 0, 0)$	$(0, 2, 0, 0, 0, 0)$	2	1	$(0, 0, -2, 0, 0, 0)$	$v_0(du_0^2 - u_1^2)$
$(0, -1, 1, 0, 0, 0)$	$(0, 0, -2, 0, 0, 0)$	1	1	$(0, -1, -1, 0, 0, 0)$	$[u_1v_0 \quad u_0v_0]$
$(0, -1, 1, 0, 0, 0)$	$(0, 0, -2, 0, 0, 0)$	2	1	$(0, -2, 0, 0, 0, 0)$	$v_0(-du_1^2 + u_0^2)$
$(0, -1, 1, 0, 0, 0)$	$(0, 2, 0, 0, 0, 0)$	1	1	$(0, 1, 1, 0, 0, 0)$	$[-u_1v_0 \quad -u_0v_0]$
$(0, -1, 1, 0, 0, 0)$	$(0, 2, 0, 0, 0, 0)$	2	1	$(0, 0, 2, 0, 0, 0)$	$v_0(-du_1^2 + u_0^2)$
$(0, 0, -2, 0, 0, 0)$	$(-1, 0, 1, 0, 0, 0)$	1	1	$(-1, 0, -1, 0, 0, 0)$	$[-u_0v_0 \quad -u_0v_1]$
$(0, 0, -2, 0, 0, 0)$	$(-1, 0, 1, 0, 0, 0)$	1	2	$(-2, 0, 0, 0, 0, 0)$	$u_0(dv_1^2 - v_0^2)$
$(0, 0, -2, 0, 0, 0)$	$(0, -1, 1, 0, 0, 0)$	1	1	$(0, -1, -1, 0, 0, 0)$	$[-u_0v_1 \quad -u_0v_0]$
$(0, 0, -2, 0, 0, 0)$	$(0, -1, 1, 0, 0, 0)$	1	2	$(0, -2, 0, 0, 0, 0)$	$u_0(dv_1^2 - v_0^2)$
$(0, 0, -2, 0, 0, 0)$	$(0, 1, 1, 0, 0, 0)$	1	1	$(0, 1, -1, 0, 0, 0)$	$[u_0v_1 \quad -u_0v_0]$
$(0, 0, -2, 0, 0, 0)$	$(0, 1, 1, 0, 0, 0)$	1	2	$(0, 2, 0, 0, 0, 0)$	$u_0(-dv_0^2 + v_1^2)$
$(0, 0, -2, 0, 0, 0)$	$(1, 0, 1, 0, 0, 0)$	1	1	$(1, 0, -1, 0, 0, 0)$	$[-u_0v_1 \quad u_0v_0]$
$(0, 0, -2, 0, 0, 0)$	$(1, 0, 1, 0, 0, 0)$	1	2	$(2, 0, 0, 0, 0, 0)$	$u_0(-dv_1^2 + v_0^2)$
$(0, 0, 2, 0, 0, 0)$	$(-1, 0, -1, 0, 0, 0)$	1	1	$(-1, 0, 1, 0, 0, 0)$	$[-u_0v_0 \quad -u_0v_1]$
$(0, 0, 2, 0, 0, 0)$	$(-1, 0, -1, 0, 0, 0)$	1	2	$(-2, 0, 0, 0, 0, 0)$	$u_0(-dv_1^2 + v_0^2)$
$(0, 0, 2, 0, 0, 0)$	$(0, -1, -1, 0, 0, 0)$	1	1	$(0, -1, 1, 0, 0, 0)$	$[-u_0v_1 \quad -u_0v_0]$
$(0, 0, 2, 0, 0, 0)$	$(0, -1, -1, 0, 0, 0)$	1	2	$(0, -2, 0, 0, 0, 0)$	$u_0(-dv_0^2 + v_1^2)$
$(0, 0, 2, 0, 0, 0)$	$(0, 1, -1, 0, 0, 0)$	1	1	$(0, 1, 1, 0, 0, 0)$	$[-u_0v_1 \quad u_0v_0]$
$(0, 0, 2, 0, 0, 0)$	$(0, 1, -1, 0, 0, 0)$	1	2	$(0, 2, 0, 0, 0, 0)$	$u_0(dv_1^2 - v_0^2)$
$(0, 0, 2, 0, 0, 0)$	$(1, 0, -1, 0, 0, 0)$	1	1	$(1, 0, 1, 0, 0, 0)$	$[u_0v_1 \quad -u_0v_0]$
$(0, 0, 2, 0, 0, 0)$	$(1, 0, -1, 0, 0, 0)$	1	2	$(2, 0, 0, 0, 0, 0)$	$u_0(dv_0^2 - v_1^2)$
$(0, 1, -1, 0, 0, 0)$	$(0, -2, 0, 0, 0, 0)$	1	1	$(0, -1, -1, 0, 0, 0)$	$[-u_1v_0 \quad u_0v_0]$
$(0, 1, -1, 0, 0, 0)$	$(0, -2, 0, 0, 0, 0)$	2	1	$(0, 0, -2, 0, 0, 0)$	$v_0(-du_1^2 + u_0^2)$
$(0, 1, -1, 0, 0, 0)$	$(0, 0, 2, 0, 0, 0)$	1	1	$(0, 1, 1, 0, 0, 0)$	$[u_1v_0 \quad -u_0v_0]$
$(0, 1, -1, 0, 0, 0)$	$(0, 0, 2, 0, 0, 0)$	2	1	$(0, 2, 0, 0, 0, 0)$	$v_0(-du_1^2 + u_0^2)$
$(0, 1, 1, 0, 0, 0)$	$(0, -2, 0, 0, 0, 0)$	1	1	$(0, -1, 1, 0, 0, 0)$	$[-u_1v_0 \quad -u_0v_0]$
$(0, 1, 1, 0, 0, 0)$	$(0, -2, 0, 0, 0, 0)$	2	1	$(0, 0, 2, 0, 0, 0)$	$v_0(du_0^2 - u_1^2)$

α	β	i	j	$i\alpha + j\beta$	$N_{ij}^{\alpha\beta}(u, v)$
(0, 1, 1, 0, 0, 0)	(0, 0, -2, 0, 0, 0)	1	1	(0, 1, -1, 0, 0, 0)	$[-u_1v_0 \quad u_0v_0]$
(0, 1, 1, 0, 0, 0)	(0, 0, -2, 0, 0, 0)	2	1	(0, 2, 0, 0, 0, 0)	$v_0(du_0^2 - u_1^2)$
(0, 2, 0, 0, 0, 0)	(-1, -1, 0, 0, 0, 0)	1	1	(-1, 1, 0, 0, 0, 0)	$[-u_0v_0 \quad -u_0v_1]$
(0, 2, 0, 0, 0, 0)	(-1, -1, 0, 0, 0, 0)	1	2	(-2, 0, 0, 0, 0, 0)	$u_0(-dv_1^2 + v_0^2)$
(0, 2, 0, 0, 0, 0)	(0, -1, -1, 0, 0, 0)	1	1	(0, 1, -1, 0, 0, 0)	$[-u_0v_1 \quad u_0v_0]$
(0, 2, 0, 0, 0, 0)	(0, -1, -1, 0, 0, 0)	1	2	(0, 0, -2, 0, 0, 0)	$u_0(-dv_0^2 + v_1^2)$
(0, 2, 0, 0, 0, 0)	(0, -1, 1, 0, 0, 0)	1	1	(0, 1, 1, 0, 0, 0)	$[u_0v_1 \quad u_0v_0]$
(0, 2, 0, 0, 0, 0)	(0, -1, 1, 0, 0, 0)	1	2	(0, 0, 2, 0, 0, 0)	$u_0(dv_1^2 - v_0^2)$
(0, 2, 0, 0, 0, 0)	(1, -1, 0, 0, 0, 0)	1	1	(1, 1, 0, 0, 0, 0)	$[u_0v_1 \quad -u_0v_0]$
(0, 2, 0, 0, 0, 0)	(1, -1, 0, 0, 0, 0)	1	2	(2, 0, 0, 0, 0, 0)	$u_0(dv_0^2 - v_1^2)$
(1, -1, 0, 0, 0, 0)	(-2, 0, 0, 0, 0, 0)	1	1	(-1, -1, 0, 0, 0, 0)	$[u_1v_0 \quad -u_0v_0]$
(1, -1, 0, 0, 0, 0)	(-2, 0, 0, 0, 0, 0)	2	1	(0, -2, 0, 0, 0, 0)	$v_0(-du_0^2 + u_1^2)$
(1, -1, 0, 0, 0, 0)	(0, 2, 0, 0, 0, 0)	1	1	(1, 1, 0, 0, 0, 0)	$[-u_1v_0 \quad u_0v_0]$
(1, -1, 0, 0, 0, 0)	(0, 2, 0, 0, 0, 0)	2	1	(2, 0, 0, 0, 0, 0)	$v_0(-du_0^2 + u_1^2)$
(1, 0, -1, 0, 0, 0)	(-2, 0, 0, 0, 0, 0)	1	1	(-1, 0, -1, 0, 0, 0)	$[u_1v_0 \quad -u_0v_0]$
(1, 0, -1, 0, 0, 0)	(-2, 0, 0, 0, 0, 0)	2	1	(0, 0, -2, 0, 0, 0)	$v_0(-du_0^2 + u_1^2)$
(1, 0, -1, 0, 0, 0)	(0, 0, 2, 0, 0, 0)	1	1	(1, 0, 1, 0, 0, 0)	$[-u_1v_0 \quad u_0v_0]$
(1, 0, -1, 0, 0, 0)	(0, 0, 2, 0, 0, 0)	2	1	(2, 0, 0, 0, 0, 0)	$v_0(-du_0^2 + u_1^2)$
(1, 0, 1, 0, 0, 0)	(-2, 0, 0, 0, 0, 0)	1	1	(-1, 0, 1, 0, 0, 0)	$[-u_0v_0 \quad -u_1v_0]$
(1, 0, 1, 0, 0, 0)	(-2, 0, 0, 0, 0, 0)	2	1	(0, 0, 2, 0, 0, 0)	$v_0(du_1^2 - u_0^2)$
(1, 0, 1, 0, 0, 0)	(0, 0, -2, 0, 0, 0)	1	1	(1, 0, -1, 0, 0, 0)	$[u_1v_0 \quad -u_0v_0]$
(1, 0, 1, 0, 0, 0)	(0, 0, -2, 0, 0, 0)	2	1	(2, 0, 0, 0, 0, 0)	$v_0(du_1^2 - u_0^2)$
(1, 1, 0, 0, 0, 0)	(-2, 0, 0, 0, 0, 0)	1	1	(-1, 1, 0, 0, 0, 0)	$[-u_0v_0 \quad -u_1v_0]$
(1, 1, 0, 0, 0, 0)	(-2, 0, 0, 0, 0, 0)	2	1	(0, 2, 0, 0, 0, 0)	$v_0(du_1^2 - u_0^2)$
(1, 1, 0, 0, 0, 0)	(0, -2, 0, 0, 0, 0)	1	1	(1, -1, 0, 0, 0, 0)	$[u_1v_0 \quad -u_0v_0]$
(1, 1, 0, 0, 0, 0)	(0, -2, 0, 0, 0, 0)	2	1	(2, 0, 0, 0, 0, 0)	$v_0(du_1^2 - u_0^2)$
(2, 0, 0, 0, 0, 0)	(-1, -1, 0, 0, 0, 0)	1	1	(1, -1, 0, 0, 0, 0)	$[u_0v_1 \quad -u_0v_0]$
(2, 0, 0, 0, 0, 0)	(-1, -1, 0, 0, 0, 0)	1	2	(0, -2, 0, 0, 0, 0)	$u_0(-dv_1^2 + v_0^2)$
(2, 0, 0, 0, 0, 0)	(-1, 0, -1, 0, 0, 0)	1	1	(1, 0, -1, 0, 0, 0)	$[u_0v_1 \quad -u_0v_0]$
(2, 0, 0, 0, 0, 0)	(-1, 0, -1, 0, 0, 0)	1	2	(0, 0, -2, 0, 0, 0)	$u_0(-dv_1^2 + v_0^2)$
(2, 0, 0, 0, 0, 0)	(-1, 0, 1, 0, 0, 0)	1	1	(1, 0, 1, 0, 0, 0)	$[u_0v_0 \quad u_0v_1]$
(2, 0, 0, 0, 0, 0)	(-1, 0, 1, 0, 0, 0)	1	2	(0, 0, 2, 0, 0, 0)	$u_0(dv_1^2 - v_0^2)$
(2, 0, 0, 0, 0, 0)	(-1, 1, 0, 0, 0, 0)	1	1	(1, 1, 0, 0, 0, 0)	$[u_0v_0 \quad u_0v_1]$
(2, 0, 0, 0, 0, 0)	(-1, 1, 0, 0, 0, 0)	1	2	(0, 2, 0, 0, 0, 0)	$u_0(dv_1^2 - v_0^2)$

5.1 Commutator identities

Below is an explicit formulation of the commutator formula identities.

CommutatorIdentitiesPlaceholder

6 Weyl group

Background theoretical info on Weyl group, especially as realized inside the group, including info on conjugation between root subgroups INCOMPLETE

6.1 Weyl group elements

The table below lists roots along with an associated Weyl group element.

Root	Weyl group element
$(-2, 0, 0, 0, 0, 0)$	$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$
$(-1, -1, 0, 0, 0, 0)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$
$(-1, 0, -1, 0, 0, 0)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
$(-1, 0, 1, 0, 0, 0)$	$\begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}$
$(-1, 1, 0, 0, 0, 0)$	$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$
$(0, -2, 0, 0, 0, 0)$	$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$
$(0, -1, -1, 0, 0, 0)$	$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \end{bmatrix}$

Root	Weyl group element
$(0, -1, 1, 0, 0, 0)$	$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$
$(0, 0, -2, 0, 0, 0)$	$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 \end{bmatrix}$
$(0, 0, 2, 0, 0, 0)$	$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 \end{bmatrix}$
$(0, 1, -1, 0, 0, 0)$	$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$
$(0, 1, 1, 0, 0, 0)$	$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \end{bmatrix}$
$(0, 2, 0, 0, 0, 0)$	$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$
$(1, -1, 0, 0, 0, 0)$	$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$
$(1, 0, -1, 0, 0, 0)$	$\begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}$

Root	Weyl group element
$(1, 0, 1, 0, 0, 0)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
$(1, 1, 0, 0, 0, 0)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$
$(2, 0, 0, 0, 0, 0)$	$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$

6.2 Weyl group conjugation coefficients

The table below lists roots along with the coefficients obtain in performing conjugation by an associated Weyl element.

Root (α)	Root (β)	$\gamma = \sigma_\alpha(\beta)$	Weyl conjugation coefficient
$(-2, 0, 0, 0, 0, 0)$	$(-2, 0, 0, 0, 0, 0)$	$(2, 0, 0, 0, 0, 0)$	$-u_0$
$(-2, 0, 0, 0, 0, 0)$	$(-1, -1, 0, 0, 0, 0)$	$(1, -1, 0, 0, 0, 0)$	$\begin{bmatrix} -u_1 & u_0 \end{bmatrix}$
$(-2, 0, 0, 0, 0, 0)$	$(-1, 0, -1, 0, 0, 0)$	$(1, 0, -1, 0, 0, 0)$	$\begin{bmatrix} -u_1 & u_0 \end{bmatrix}$
$(-2, 0, 0, 0, 0, 0)$	$(-1, 0, 1, 0, 0, 0)$	$(1, 0, 1, 0, 0, 0)$	$\begin{bmatrix} -u_0 & -u_1 \end{bmatrix}$
$(-2, 0, 0, 0, 0, 0)$	$(-1, 1, 0, 0, 0, 0)$	$(1, 1, 0, 0, 0, 0)$	$\begin{bmatrix} -u_0 & -u_1 \end{bmatrix}$
$(-2, 0, 0, 0, 0, 0)$	$(0, -2, 0, 0, 0, 0)$	$(0, -2, 0, 0, 0, 0)$	u_0
$(-2, 0, 0, 0, 0, 0)$	$(0, -1, -1, 0, 0, 0)$	$(0, -1, -1, 0, 0, 0)$	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
$(-2, 0, 0, 0, 0, 0)$	$(0, -1, 1, 0, 0, 0)$	$(0, -1, 1, 0, 0, 0)$	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
$(-2, 0, 0, 0, 0, 0)$	$(0, 0, -2, 0, 0, 0)$	$(0, 0, -2, 0, 0, 0)$	u_0
$(-2, 0, 0, 0, 0, 0)$	$(0, 0, 2, 0, 0, 0)$	$(0, 0, 2, 0, 0, 0)$	u_0
$(-2, 0, 0, 0, 0, 0)$	$(0, 1, -1, 0, 0, 0)$	$(0, 1, -1, 0, 0, 0)$	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
$(-2, 0, 0, 0, 0, 0)$	$(0, 1, 1, 0, 0, 0)$	$(0, 1, 1, 0, 0, 0)$	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
$(-2, 0, 0, 0, 0, 0)$	$(0, 2, 0, 0, 0, 0)$	$(0, 2, 0, 0, 0, 0)$	u_0
$(-2, 0, 0, 0, 0, 0)$	$(1, -1, 0, 0, 0, 0)$	$(-1, -1, 0, 0, 0, 0)$	$\begin{bmatrix} -u_1 & u_0 \end{bmatrix}$
$(-2, 0, 0, 0, 0, 0)$	$(1, 0, -1, 0, 0, 0)$	$(-1, 0, -1, 0, 0, 0)$	$\begin{bmatrix} -u_1 & u_0 \end{bmatrix}$
$(-2, 0, 0, 0, 0, 0)$	$(1, 0, 1, 0, 0, 0)$	$(-1, 0, 1, 0, 0, 0)$	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
$(-2, 0, 0, 0, 0, 0)$	$(1, 1, 0, 0, 0, 0)$	$(-1, 1, 0, 0, 0, 0)$	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
$(-2, 0, 0, 0, 0, 0)$	$(2, 0, 0, 0, 0, 0)$	$(-2, 0, 0, 0, 0, 0)$	$-u_0$
$(-1, -1, 0, 0, 0, 0)$	$(-2, 0, 0, 0, 0, 0)$	$(0, 2, 0, 0, 0, 0)$	$-u_0$
$(-1, -1, 0, 0, 0, 0)$	$(-1, -1, 0, 0, 0, 0)$	$(1, 1, 0, 0, 0, 0)$	$\begin{bmatrix} -u_0 & u_1 \end{bmatrix}$
$(-1, -1, 0, 0, 0, 0)$	$(-1, 0, -1, 0, 0, 0)$	$(0, 1, -1, 0, 0, 0)$	$\begin{bmatrix} u_0 & -u_1 \end{bmatrix}$
$(-1, -1, 0, 0, 0, 0)$	$(-1, 0, 1, 0, 0, 0)$	$(0, 1, 1, 0, 0, 0)$	$\begin{bmatrix} -u_1 & -u_0 \end{bmatrix}$
$(-1, -1, 0, 0, 0, 0)$	$(-1, 1, 0, 0, 0, 0)$	$(-1, 1, 0, 0, 0, 0)$	$\begin{bmatrix} -u_0 & u_1 \end{bmatrix}$
$(-1, -1, 0, 0, 0, 0)$	$(0, -2, 0, 0, 0, 0)$	$(2, 0, 0, 0, 0, 0)$	$-u_0$
$(-1, -1, 0, 0, 0, 0)$	$(0, -1, -1, 0, 0, 0)$	$(1, 0, -1, 0, 0, 0)$	$\begin{bmatrix} -u_0 & u_1 \end{bmatrix}$
$(-1, -1, 0, 0, 0, 0)$	$(0, -1, 1, 0, 0, 0)$	$(1, 0, 1, 0, 0, 0)$	$\begin{bmatrix} -u_0 & -u_1 \end{bmatrix}$

Root (α)	Root (β)	$\gamma = \sigma_\alpha(\beta)$	Weyl conjugation coefficient
$(-1, -1, 0, 0, 0, 0)$	$(0, 0, -2, 0, 0, 0)$	$(0, 0, -2, 0, 0, 0)$	u_0
$(-1, -1, 0, 0, 0, 0)$	$(0, 0, 2, 0, 0, 0)$	$(0, 0, 2, 0, 0, 0)$	u_0
$(-1, -1, 0, 0, 0, 0)$	$(0, 1, -1, 0, 0, 0)$	$(-1, 0, -1, 0, 0, 0)$	$-u_0 \quad u_1$
$(-1, -1, 0, 0, 0, 0)$	$(0, 1, 1, 0, 0, 0)$	$(-1, 0, 1, 0, 0, 0)$	$u_1 \quad u_0$
$(-1, -1, 0, 0, 0, 0)$	$(0, 2, 0, 0, 0, 0)$	$(-2, 0, 0, 0, 0, 0)$	$-u_0$
$(-1, -1, 0, 0, 0, 0)$	$(1, -1, 0, 0, 0, 0)$	$(1, -1, 0, 0, 0, 0)$	$u_0 \quad -u_1$
$(-1, -1, 0, 0, 0, 0)$	$(1, 0, -1, 0, 0, 0)$	$(0, -1, -1, 0, 0, 0)$	$u_0 \quad -u_1$
$(-1, -1, 0, 0, 0, 0)$	$(1, 0, 1, 0, 0, 0)$	$(0, -1, 1, 0, 0, 0)$	$u_0 \quad u_1$
$(-1, -1, 0, 0, 0, 0)$	$(1, 1, 0, 0, 0, 0)$	$(-1, -1, 0, 0, 0, 0)$	$-u_0 \quad u_1$
$(-1, -1, 0, 0, 0, 0)$	$(2, 0, 0, 0, 0, 0)$	$(0, -2, 0, 0, 0, 0)$	$-u_0$
$(-1, 0, -1, 0, 0, 0)$	$(-2, 0, 0, 0, 0, 0)$	$(0, 0, 2, 0, 0, 0)$	$-u_0$
$(-1, 0, -1, 0, 0, 0)$	$(-1, -1, 0, 0, 0, 0)$	$(0, -1, 1, 0, 0, 0)$	$u_0 \quad -u_1$
$(-1, 0, -1, 0, 0, 0)$	$(-1, 0, -1, 0, 0, 0)$	$(1, 0, 1, 0, 0, 0)$	$-u_0 \quad u_1$
$(-1, 0, -1, 0, 0, 0)$	$(-1, 0, 1, 0, 0, 0)$	$(-1, 0, 1, 0, 0, 0)$	$-u_0 \quad u_1$
$(-1, 0, -1, 0, 0, 0)$	$(-1, 1, 0, 0, 0, 0)$	$(0, 1, 1, 0, 0, 0)$	$u_1 \quad -u_0$
$(-1, 0, -1, 0, 0, 0)$	$(0, -2, 0, 0, 0, 0)$	$(0, -2, 0, 0, 0, 0)$	u_0
$(-1, 0, -1, 0, 0, 0)$	$(0, -1, -1, 0, 0, 0)$	$(1, -1, 0, 0, 0, 0)$	$u_0 \quad u_1$
$(-1, 0, -1, 0, 0, 0)$	$(0, -1, 1, 0, 0, 0)$	$(-1, -1, 0, 0, 0, 0)$	$-u_0 \quad u_1$
$(-1, 0, -1, 0, 0, 0)$	$(0, 0, -2, 0, 0, 0)$	$(2, 0, 0, 0, 0, 0)$	$-u_0$
$(-1, 0, -1, 0, 0, 0)$	$(0, 0, 2, 0, 0, 0)$	$(-2, 0, 0, 0, 0, 0)$	$-u_0$
$(-1, 0, -1, 0, 0, 0)$	$(0, 1, -1, 0, 0, 0)$	$(1, 1, 0, 0, 0, 0)$	$-u_0 \quad -u_1$
$(-1, 0, -1, 0, 0, 0)$	$(0, 1, 1, 0, 0, 0)$	$(-1, 1, 0, 0, 0, 0)$	$u_1 \quad -u_0$
$(-1, 0, -1, 0, 0, 0)$	$(0, 2, 0, 0, 0, 0)$	$(0, 2, 0, 0, 0, 0)$	u_0
$(-1, 0, -1, 0, 0, 0)$	$(1, -1, 0, 0, 0, 0)$	$(0, -1, -1, 0, 0, 0)$	$-u_0 \quad -u_1$
$(-1, 0, -1, 0, 0, 0)$	$(1, 0, -1, 0, 0, 0)$	$(1, 0, -1, 0, 0, 0)$	$u_0 \quad -u_1$
$(-1, 0, -1, 0, 0, 0)$	$(1, 0, 1, 0, 0, 0)$	$(-1, 0, -1, 0, 0, 0)$	$-u_0 \quad u_1$
$(-1, 0, -1, 0, 0, 0)$	$(1, 1, 0, 0, 0, 0)$	$(0, 1, -1, 0, 0, 0)$	$u_0 \quad u_1$
$(-1, 0, -1, 0, 0, 0)$	$(2, 0, 0, 0, 0, 0)$	$(0, 0, -2, 0, 0, 0)$	$-u_0$
$(-1, 0, 1, 0, 0, 0)$	$(-2, 0, 0, 0, 0, 0)$	$(0, 0, -2, 0, 0, 0)$	u_0
$(-1, 0, 1, 0, 0, 0)$	$(-1, -1, 0, 0, 0, 0)$	$(0, -1, -1, 0, 0, 0)$	$-u_1 \quad u_0$
$(-1, 0, 1, 0, 0, 0)$	$(-1, 0, -1, 0, 0, 0)$	$(-1, 0, -1, 0, 0, 0)$	$u_0 \quad -u_1$
$(-1, 0, 1, 0, 0, 0)$	$(-1, 0, 1, 0, 0, 0)$	$(1, 0, -1, 0, 0, 0)$	$u_1 \quad u_0$
$(-1, 0, 1, 0, 0, 0)$	$(-1, 1, 0, 0, 0, 0)$	$(0, 1, -1, 0, 0, 0)$	$u_0 \quad u_1$
$(-1, 0, 1, 0, 0, 0)$	$(0, -2, 0, 0, 0, 0)$	$(0, -2, 0, 0, 0, 0)$	u_0
$(-1, 0, 1, 0, 0, 0)$	$(0, -1, -1, 0, 0, 0)$	$(-1, -1, 0, 0, 0, 0)$	$u_1 \quad -u_0$
$(-1, 0, 1, 0, 0, 0)$	$(0, -1, 1, 0, 0, 0)$	$(1, -1, 0, 0, 0, 0)$	$u_1 \quad u_0$
$(-1, 0, 1, 0, 0, 0)$	$(0, 0, -2, 0, 0, 0)$	$(-2, 0, 0, 0, 0, 0)$	u_0
$(-1, 0, 1, 0, 0, 0)$	$(0, 0, 2, 0, 0, 0)$	$(2, 0, 0, 0, 0, 0)$	u_0
$(-1, 0, 1, 0, 0, 0)$	$(0, 1, -1, 0, 0, 0)$	$(-1, 1, 0, 0, 0, 0)$	$u_0 \quad u_1$
$(-1, 0, 1, 0, 0, 0)$	$(0, 1, 1, 0, 0, 0)$	$(1, 1, 0, 0, 0, 0)$	$u_1 \quad -u_0$
$(-1, 0, 1, 0, 0, 0)$	$(0, 2, 0, 0, 0, 0)$	$(0, 2, 0, 0, 0, 0)$	u_0
$(-1, 0, 1, 0, 0, 0)$	$(1, -1, 0, 0, 0, 0)$	$(0, -1, 1, 0, 0, 0)$	$u_1 \quad u_0$
$(-1, 0, 1, 0, 0, 0)$	$(1, 0, -1, 0, 0, 0)$	$(-1, 0, 1, 0, 0, 0)$	$u_1 \quad u_0$
$(-1, 0, 1, 0, 0, 0)$	$(1, 0, 1, 0, 0, 0)$	$(1, 0, 1, 0, 0, 0)$	$u_0 \quad -u_1$
$(-1, 0, 1, 0, 0, 0)$	$(1, 1, 0, 0, 0, 0)$	$(0, 1, 1, 0, 0, 0)$	$-u_1 \quad u_0$
$(-1, 0, 1, 0, 0, 0)$	$(2, 0, 0, 0, 0, 0)$	$(0, 0, 2, 0, 0, 0)$	u_0
$(-1, 1, 0, 0, 0, 0)$	$(-2, 0, 0, 0, 0, 0)$	$(0, -2, 0, 0, 0, 0)$	u_0
$(-1, 1, 0, 0, 0, 0)$	$(-1, -1, 0, 0, 0, 0)$	$(-1, -1, 0, 0, 0, 0)$	$u_0 \quad -u_1$
$(-1, 1, 0, 0, 0, 0)$	$(-1, 0, -1, 0, 0, 0)$	$(0, -1, -1, 0, 0, 0)$	$u_1 \quad u_0$
$(-1, 1, 0, 0, 0, 0)$	$(-1, 0, 1, 0, 0, 0)$	$(0, -1, 1, 0, 0, 0)$	$u_0 \quad u_1$

Root (α)	Root (β)	$\gamma = \sigma_\alpha(\beta)$	Weyl conjugation coefficient
$(-1, 1, 0, 0, 0, 0)$	$(-1, 1, 0, 0, 0, 0)$	$(1, -1, 0, 0, 0, 0)$	$\begin{bmatrix} u_1 & u_0 \end{bmatrix}$
$(-1, 1, 0, 0, 0, 0)$	$(0, -2, 0, 0, 0, 0)$	$(-2, 0, 0, 0, 0, 0)$	u_0
$(-1, 1, 0, 0, 0, 0)$	$(0, -1, -1, 0, 0, 0)$	$(-1, 0, -1, 0, 0, 0)$	$\begin{bmatrix} u_1 & u_0 \end{bmatrix}$
$(-1, 1, 0, 0, 0, 0)$	$(0, -1, 1, 0, 0, 0)$	$(-1, 0, 1, 0, 0, 0)$	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
$(-1, 1, 0, 0, 0, 0)$	$(0, 0, -2, 0, 0, 0)$	$(0, 0, -2, 0, 0, 0)$	u_0
$(-1, 1, 0, 0, 0, 0)$	$(0, 0, 2, 0, 0, 0)$	$(0, 0, 2, 0, 0, 0)$	u_0
$(-1, 1, 0, 0, 0, 0)$	$(0, 1, -1, 0, 0, 0)$	$(1, 0, -1, 0, 0, 0)$	$\begin{bmatrix} u_1 & u_0 \end{bmatrix}$
$(-1, 1, 0, 0, 0, 0)$	$(0, 1, 1, 0, 0, 0)$	$(1, 0, 1, 0, 0, 0)$	$\begin{bmatrix} u_1 & u_0 \end{bmatrix}$
$(-1, 1, 0, 0, 0, 0)$	$(0, 2, 0, 0, 0, 0)$	$(2, 0, 0, 0, 0, 0)$	u_0
$(-1, 1, 0, 0, 0, 0)$	$(1, -1, 0, 0, 0, 0)$	$(-1, 1, 0, 0, 0, 0)$	$\begin{bmatrix} u_1 & u_0 \end{bmatrix}$
$(-1, 1, 0, 0, 0, 0)$	$(1, 0, -1, 0, 0, 0)$	$(0, 1, -1, 0, 0, 0)$	$\begin{bmatrix} u_1 & u_0 \end{bmatrix}$
$(-1, 1, 0, 0, 0, 0)$	$(1, 0, 1, 0, 0, 0)$	$(0, 1, 1, 0, 0, 0)$	$\begin{bmatrix} u_1 & u_0 \end{bmatrix}$
$(-1, 1, 0, 0, 0, 0)$	$(1, 1, 0, 0, 0, 0)$	$(1, 1, 0, 0, 0, 0)$	$\begin{bmatrix} u_0 & -u_1 \end{bmatrix}$
$(-1, 1, 0, 0, 0, 0)$	$(2, 0, 0, 0, 0, 0)$	$(0, 2, 0, 0, 0, 0)$	u_0
$(0, -2, 0, 0, 0, 0)$	$(-2, 0, 0, 0, 0, 0)$	$(-2, 0, 0, 0, 0, 0)$	u_0
$(0, -2, 0, 0, 0, 0)$	$(-1, -1, 0, 0, 0, 0)$	$(-1, 1, 0, 0, 0, 0)$	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
$(0, -2, 0, 0, 0, 0)$	$(-1, 0, -1, 0, 0, 0)$	$(-1, 0, -1, 0, 0, 0)$	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
$(0, -2, 0, 0, 0, 0)$	$(-1, 0, 1, 0, 0, 0)$	$(-1, 0, 1, 0, 0, 0)$	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
$(0, -2, 0, 0, 0, 0)$	$(-1, 1, 0, 0, 0, 0)$	$(-1, -1, 0, 0, 0, 0)$	$\begin{bmatrix} -u_0 & -u_1 \end{bmatrix}$
$(0, -2, 0, 0, 0, 0)$	$(0, -2, 0, 0, 0, 0)$	$(0, 2, 0, 0, 0, 0)$	$-u_0$
$(0, -2, 0, 0, 0, 0)$	$(0, -1, -1, 0, 0, 0)$	$(0, 1, -1, 0, 0, 0)$	$\begin{bmatrix} u_1 & -u_0 \end{bmatrix}$
$(0, -2, 0, 0, 0, 0)$	$(0, -1, 1, 0, 0, 0)$	$(0, 1, 1, 0, 0, 0)$	$\begin{bmatrix} -u_1 & -u_0 \end{bmatrix}$
$(0, -2, 0, 0, 0, 0)$	$(0, 0, -2, 0, 0, 0)$	$(0, 0, -2, 0, 0, 0)$	u_0
$(0, -2, 0, 0, 0, 0)$	$(0, 0, 2, 0, 0, 0)$	$(0, 0, 2, 0, 0, 0)$	u_0
$(0, -2, 0, 0, 0, 0)$	$(0, 1, -1, 0, 0, 0)$	$(0, -1, -1, 0, 0, 0)$	$\begin{bmatrix} u_1 & -u_0 \end{bmatrix}$
$(0, -2, 0, 0, 0, 0)$	$(0, 1, 1, 0, 0, 0)$	$(0, -1, 1, 0, 0, 0)$	$\begin{bmatrix} u_1 & u_0 \end{bmatrix}$
$(0, -2, 0, 0, 0, 0)$	$(0, 2, 0, 0, 0, 0)$	$(0, -2, 0, 0, 0, 0)$	$-u_0$
$(0, -2, 0, 0, 0, 0)$	$(1, -1, 0, 0, 0, 0)$	$(1, 1, 0, 0, 0, 0)$	$\begin{bmatrix} -u_1 & u_0 \end{bmatrix}$
$(0, -2, 0, 0, 0, 0)$	$(1, 0, -1, 0, 0, 0)$	$(1, 0, -1, 0, 0, 0)$	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
$(0, -2, 0, 0, 0, 0)$	$(1, 0, 1, 0, 0, 0)$	$(1, 0, 1, 0, 0, 0)$	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
$(0, -2, 0, 0, 0, 0)$	$(1, 1, 0, 0, 0, 0)$	$(1, -1, 0, 0, 0, 0)$	$\begin{bmatrix} -u_1 & u_0 \end{bmatrix}$
$(0, -2, 0, 0, 0, 0)$	$(2, 0, 0, 0, 0, 0)$	$(2, 0, 0, 0, 0, 0)$	u_0
$(0, -1, -1, 0, 0, 0)$	$(-2, 0, 0, 0, 0, 0)$	$(-2, 0, 0, 0, 0, 0)$	u_0
$(0, -1, -1, 0, 0, 0)$	$(-1, -1, 0, 0, 0, 0)$	$(-1, 0, 1, 0, 0, 0)$	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
$(0, -1, -1, 0, 0, 0)$	$(-1, 0, -1, 0, 0, 0)$	$(-1, 1, 0, 0, 0, 0)$	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
$(0, -1, -1, 0, 0, 0)$	$(-1, 0, 1, 0, 0, 0)$	$(-1, -1, 0, 0, 0, 0)$	$\begin{bmatrix} -u_0 & -u_1 \end{bmatrix}$
$(0, -1, -1, 0, 0, 0)$	$(-1, 1, 0, 0, 0, 0)$	$(-1, 0, -1, 0, 0, 0)$	$\begin{bmatrix} -u_0 & -u_1 \end{bmatrix}$
$(0, -1, -1, 0, 0, 0)$	$(0, -2, 0, 0, 0, 0)$	$(0, 0, 2, 0, 0, 0)$	$-u_0$
$(0, -1, -1, 0, 0, 0)$	$(0, -1, -1, 0, 0, 0)$	$(0, 1, 1, 0, 0, 0)$	$\begin{bmatrix} u_0 & -u_1 \end{bmatrix}$
$(0, -1, -1, 0, 0, 0)$	$(0, -1, 1, 0, 0, 0)$	$(0, -1, 1, 0, 0, 0)$	$\begin{bmatrix} -u_0 & u_1 \end{bmatrix}$
$(0, -1, -1, 0, 0, 0)$	$(0, 0, -2, 0, 0, 0)$	$(0, 2, 0, 0, 0, 0)$	$-u_0$
$(0, -1, -1, 0, 0, 0)$	$(0, 0, 2, 0, 0, 0)$	$(0, -2, 0, 0, 0, 0)$	$-u_0$
$(0, -1, -1, 0, 0, 0)$	$(0, 1, -1, 0, 0, 0)$	$(0, 1, -1, 0, 0, 0)$	$\begin{bmatrix} -u_0 & u_1 \end{bmatrix}$
$(0, -1, -1, 0, 0, 0)$	$(0, 1, 1, 0, 0, 0)$	$(0, -1, -1, 0, 0, 0)$	$\begin{bmatrix} u_0 & -u_1 \end{bmatrix}$
$(0, -1, -1, 0, 0, 0)$	$(0, 2, 0, 0, 0, 0)$	$(0, 0, -2, 0, 0, 0)$	$-u_0$
$(0, -1, -1, 0, 0, 0)$	$(1, -1, 0, 0, 0, 0)$	$(1, 0, 1, 0, 0, 0)$	$\begin{bmatrix} -u_1 & u_0 \end{bmatrix}$
$(0, -1, -1, 0, 0, 0)$	$(1, 0, -1, 0, 0, 0)$	$(1, 1, 0, 0, 0, 0)$	$\begin{bmatrix} -u_1 & u_0 \end{bmatrix}$
$(0, -1, -1, 0, 0, 0)$	$(1, 0, 1, 0, 0, 0)$	$(1, -1, 0, 0, 0, 0)$	$\begin{bmatrix} -u_1 & u_0 \end{bmatrix}$
$(0, -1, -1, 0, 0, 0)$	$(1, 1, 0, 0, 0, 0)$	$(1, 0, -1, 0, 0, 0)$	$\begin{bmatrix} -u_1 & u_0 \end{bmatrix}$
$(0, -1, -1, 0, 0, 0)$	$(2, 0, 0, 0, 0, 0)$	$(2, 0, 0, 0, 0, 0)$	u_0

Root (α)	Root (β)	$\gamma = \sigma_\alpha(\beta)$	Weyl conjugation coefficient
(0, -1, 1, 0, 0, 0)	(-2, 0, 0, 0, 0, 0)	(-2, 0, 0, 0, 0, 0)	u_0
(0, -1, 1, 0, 0, 0)	(-1, -1, 0, 0, 0, 0)	(-1, 0, -1, 0, 0, 0)	$u_0 \quad u_1$
(0, -1, 1, 0, 0, 0)	(-1, 0, -1, 0, 0, 0)	(-1, -1, 0, 0, 0, 0)	$u_0 \quad u_1$
(0, -1, 1, 0, 0, 0)	(-1, 0, 1, 0, 0, 0)	(-1, 1, 0, 0, 0, 0)	$u_0 \quad u_1$
(0, -1, 1, 0, 0, 0)	(-1, 1, 0, 0, 0, 0)	(-1, 0, 1, 0, 0, 0)	$u_0 \quad u_1$
(0, -1, 1, 0, 0, 0)	(0, -2, 0, 0, 0, 0)	(0, 0, -2, 0, 0, 0)	u_0
(0, -1, 1, 0, 0, 0)	(0, -1, -1, 0, 0, 0)	(0, -1, -1, 0, 0, 0)	$-u_0 \quad u_1$
(0, -1, 1, 0, 0, 0)	(0, -1, 1, 0, 0, 0)	(0, 1, -1, 0, 0, 0)	$u_0 \quad u_1$
(0, -1, 1, 0, 0, 0)	(0, 0, -2, 0, 0, 0)	(0, -2, 0, 0, 0, 0)	u_0
(0, -1, 1, 0, 0, 0)	(0, 0, 2, 0, 0, 0)	(0, 2, 0, 0, 0, 0)	u_0
(0, -1, 1, 0, 0, 0)	(0, 1, -1, 0, 0, 0)	(0, -1, 1, 0, 0, 0)	$u_0 \quad u_1$
(0, -1, 1, 0, 0, 0)	(0, 1, 1, 0, 0, 0)	(0, 1, 1, 0, 0, 0)	$-u_0 \quad u_1$
(0, -1, 1, 0, 0, 0)	(0, 2, 0, 0, 0, 0)	(0, 0, 2, 0, 0, 0)	u_0
(0, -1, 1, 0, 0, 0)	(1, -1, 0, 0, 0, 0)	(1, 0, -1, 0, 0, 0)	$u_0 \quad u_1$
(0, -1, 1, 0, 0, 0)	(1, 0, -1, 0, 0, 0)	(1, -1, 0, 0, 0, 0)	$u_0 \quad u_1$
(0, -1, 1, 0, 0, 0)	(1, 0, 1, 0, 0, 0)	(1, 1, 0, 0, 0, 0)	$u_0 \quad u_1$
(0, -1, 1, 0, 0, 0)	(1, 1, 0, 0, 0, 0)	(1, 0, 1, 0, 0, 0)	$u_0 \quad u_1$
(0, -1, 1, 0, 0, 0)	(2, 0, 0, 0, 0, 0)	(2, 0, 0, 0, 0, 0)	u_0
(0, 0, -2, 0, 0, 0)	(-2, 0, 0, 0, 0, 0)	(-2, 0, 0, 0, 0, 0)	u_0
(0, 0, -2, 0, 0, 0)	(-1, -1, 0, 0, 0, 0)	(-1, -1, 0, 0, 0, 0)	$u_0 \quad u_1$
(0, 0, -2, 0, 0, 0)	(-1, 0, -1, 0, 0, 0)	(-1, 0, 1, 0, 0, 0)	$u_0 \quad u_1$
(0, 0, -2, 0, 0, 0)	(-1, 0, 1, 0, 0, 0)	(-1, 0, -1, 0, 0, 0)	$-u_0 \quad -u_1$
(0, 0, -2, 0, 0, 0)	(-1, 1, 0, 0, 0, 0)	(-1, 1, 0, 0, 0, 0)	$u_0 \quad u_1$
(0, 0, -2, 0, 0, 0)	(0, -2, 0, 0, 0, 0)	(0, -2, 0, 0, 0, 0)	u_0
(0, 0, -2, 0, 0, 0)	(0, -1, -1, 0, 0, 0)	(0, -1, 1, 0, 0, 0)	$u_1 \quad u_0$
(0, 0, -2, 0, 0, 0)	(0, -1, 1, 0, 0, 0)	(0, -1, -1, 0, 0, 0)	$-u_1 \quad -u_0$
(0, 0, -2, 0, 0, 0)	(0, 0, -2, 0, 0, 0)	(0, 0, 2, 0, 0, 0)	$-u_0$
(0, 0, -2, 0, 0, 0)	(0, 0, 2, 0, 0, 0)	(0, 0, -2, 0, 0, 0)	$-u_0$
(0, 0, -2, 0, 0, 0)	(0, 1, -1, 0, 0, 0)	(0, 1, 1, 0, 0, 0)	$u_1 \quad -u_0$
(0, 0, -2, 0, 0, 0)	(0, 1, 1, 0, 0, 0)	(0, 1, -1, 0, 0, 0)	$u_1 \quad -u_0$
(0, 0, -2, 0, 0, 0)	(0, 2, 0, 0, 0, 0)	(0, 2, 0, 0, 0, 0)	u_0
(0, 0, -2, 0, 0, 0)	(1, -1, 0, 0, 0, 0)	(1, -1, 0, 0, 0, 0)	$u_0 \quad u_1$
(0, 0, -2, 0, 0, 0)	(1, 0, -1, 0, 0, 0)	(1, 0, 1, 0, 0, 0)	$-u_1 \quad u_0$
(0, 0, -2, 0, 0, 0)	(1, 0, 1, 0, 0, 0)	(1, 0, -1, 0, 0, 0)	$-u_1 \quad u_0$
(0, 0, -2, 0, 0, 0)	(1, 1, 0, 0, 0, 0)	(1, 1, 0, 0, 0, 0)	$u_0 \quad u_1$
(0, 0, -2, 0, 0, 0)	(2, 0, 0, 0, 0, 0)	(2, 0, 0, 0, 0, 0)	u_0
(0, 0, 2, 0, 0, 0)	(-2, 0, 0, 0, 0, 0)	(-2, 0, 0, 0, 0, 0)	u_0
(0, 0, 2, 0, 0, 0)	(-1, -1, 0, 0, 0, 0)	(-1, -1, 0, 0, 0, 0)	$u_0 \quad u_1$
(0, 0, 2, 0, 0, 0)	(-1, 0, -1, 0, 0, 0)	(-1, 0, 1, 0, 0, 0)	$u_0 \quad u_1$
(0, 0, 2, 0, 0, 0)	(-1, 0, 1, 0, 0, 0)	(-1, 0, -1, 0, 0, 0)	$-u_0 \quad -u_1$
(0, 0, 2, 0, 0, 0)	(-1, 1, 0, 0, 0, 0)	(-1, 1, 0, 0, 0, 0)	$u_0 \quad u_1$
(0, 0, 2, 0, 0, 0)	(0, -2, 0, 0, 0, 0)	(0, -2, 0, 0, 0, 0)	u_0
(0, 0, 2, 0, 0, 0)	(0, -1, -1, 0, 0, 0)	(0, -1, 1, 0, 0, 0)	$u_1 \quad u_0$
(0, 0, 2, 0, 0, 0)	(0, -1, 1, 0, 0, 0)	(0, -1, -1, 0, 0, 0)	$-u_1 \quad -u_0$
(0, 0, 2, 0, 0, 0)	(0, 0, -2, 0, 0, 0)	(0, 0, 2, 0, 0, 0)	$-u_0$
(0, 0, 2, 0, 0, 0)	(0, 0, 2, 0, 0, 0)	(0, 0, -2, 0, 0, 0)	$-u_0$
(0, 0, 2, 0, 0, 0)	(0, 1, -1, 0, 0, 0)	(0, 1, 1, 0, 0, 0)	$u_1 \quad -u_0$
(0, 0, 2, 0, 0, 0)	(0, 1, 1, 0, 0, 0)	(0, 1, -1, 0, 0, 0)	

Root (α)	Root (β)	$\gamma = \sigma_\alpha(\beta)$	Weyl conjugation coefficient
(0, 0, 2, 0, 0, 0)	(1, 0, -1, 0, 0, 0)	(1, 0, 1, 0, 0, 0)	$\begin{bmatrix} -u_1 & u_0 \end{bmatrix}$
(0, 0, 2, 0, 0, 0)	(1, 0, 1, 0, 0, 0)	(1, 0, -1, 0, 0, 0)	$\begin{bmatrix} -u_1 & u_0 \end{bmatrix}$
(0, 0, 2, 0, 0, 0)	(1, 1, 0, 0, 0, 0)	(1, 1, 0, 0, 0, 0)	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
(0, 0, 2, 0, 0, 0)	(2, 0, 0, 0, 0, 0)	(2, 0, 0, 0, 0, 0)	u_0
(0, 1, -1, 0, 0, 0)	(-2, 0, 0, 0, 0, 0)	(-2, 0, 0, 0, 0, 0)	u_0
(0, 1, -1, 0, 0, 0)	(-1, -1, 0, 0, 0, 0)	(-1, 0, -1, 0, 0, 0)	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
(0, 1, -1, 0, 0, 0)	(-1, 0, -1, 0, 0, 0)	(-1, -1, 0, 0, 0, 0)	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
(0, 1, -1, 0, 0, 0)	(-1, 0, 1, 0, 0, 0)	(-1, 1, 0, 0, 0, 0)	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
(0, 1, -1, 0, 0, 0)	(-1, 1, 0, 0, 0, 0)	(-1, 0, 1, 0, 0, 0)	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
(0, 1, -1, 0, 0, 0)	(0, -2, 0, 0, 0, 0)	(0, 0, -2, 0, 0, 0)	u_0
(0, 1, -1, 0, 0, 0)	(0, -1, -1, 0, 0, 0)	(0, -1, -1, 0, 0, 0)	$\begin{bmatrix} -u_0 & u_1 \end{bmatrix}$
(0, 1, -1, 0, 0, 0)	(0, -1, 1, 0, 0, 0)	(0, 1, -1, 0, 0, 0)	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
(0, 1, -1, 0, 0, 0)	(0, 0, -2, 0, 0, 0)	(0, -2, 0, 0, 0, 0)	u_0
(0, 1, -1, 0, 0, 0)	(0, 0, 2, 0, 0, 0)	(0, 2, 0, 0, 0, 0)	u_0
(0, 1, -1, 0, 0, 0)	(0, 1, -1, 0, 0, 0)	(0, -1, 1, 0, 0, 0)	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
(0, 1, -1, 0, 0, 0)	(0, 1, 1, 0, 0, 0)	(0, 1, 1, 0, 0, 0)	$\begin{bmatrix} -u_0 & u_1 \end{bmatrix}$
(0, 1, -1, 0, 0, 0)	(0, 2, 0, 0, 0, 0)	(0, 0, 2, 0, 0, 0)	u_0
(0, 1, -1, 0, 0, 0)	(1, -1, 0, 0, 0, 0)	(1, 0, -1, 0, 0, 0)	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
(0, 1, -1, 0, 0, 0)	(1, 0, -1, 0, 0, 0)	(1, -1, 0, 0, 0, 0)	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
(0, 1, -1, 0, 0, 0)	(1, 0, 1, 0, 0, 0)	(1, 1, 0, 0, 0, 0)	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
(0, 1, -1, 0, 0, 0)	(1, 1, 0, 0, 0, 0)	(1, 0, 1, 0, 0, 0)	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
(0, 1, -1, 0, 0, 0)	(2, 0, 0, 0, 0, 0)	(2, 0, 0, 0, 0, 0)	u_0
(0, 1, 1, 0, 0, 0)	(-2, 0, 0, 0, 0, 0)	(-2, 0, 0, 0, 0, 0)	u_0
(0, 1, 1, 0, 0, 0)	(-1, -1, 0, 0, 0, 0)	(-1, 0, 1, 0, 0, 0)	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
(0, 1, 1, 0, 0, 0)	(-1, 0, -1, 0, 0, 0)	(-1, 1, 0, 0, 0, 0)	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
(0, 1, 1, 0, 0, 0)	(-1, 0, 1, 0, 0, 0)	(-1, -1, 0, 0, 0, 0)	$\begin{bmatrix} -u_0 & -u_1 \end{bmatrix}$
(0, 1, 1, 0, 0, 0)	(-1, 1, 0, 0, 0, 0)	(-1, 0, -1, 0, 0, 0)	$\begin{bmatrix} -u_0 & -u_1 \end{bmatrix}$
(0, 1, 1, 0, 0, 0)	(0, -2, 0, 0, 0, 0)	(0, 0, 2, 0, 0, 0)	$-u_0$
(0, 1, 1, 0, 0, 0)	(0, -1, -1, 0, 0, 0)	(0, 1, 1, 0, 0, 0)	$\begin{bmatrix} u_0 & -u_1 \end{bmatrix}$
(0, 1, 1, 0, 0, 0)	(0, -1, 1, 0, 0, 0)	(0, -1, 1, 0, 0, 0)	$\begin{bmatrix} -u_0 & u_1 \end{bmatrix}$
(0, 1, 1, 0, 0, 0)	(0, 0, -2, 0, 0, 0)	(0, 2, 0, 0, 0, 0)	$-u_0$
(0, 1, 1, 0, 0, 0)	(0, 0, 2, 0, 0, 0)	(0, -2, 0, 0, 0, 0)	$-u_0$
(0, 1, 1, 0, 0, 0)	(0, 1, -1, 0, 0, 0)	(0, 1, -1, 0, 0, 0)	$\begin{bmatrix} -u_0 & u_1 \end{bmatrix}$
(0, 1, 1, 0, 0, 0)	(0, 1, 1, 0, 0, 0)	(0, -1, -1, 0, 0, 0)	$\begin{bmatrix} u_0 & -u_1 \end{bmatrix}$
(0, 1, 1, 0, 0, 0)	(0, 2, 0, 0, 0, 0)	(0, 0, -2, 0, 0, 0)	$-u_0$
(0, 1, 1, 0, 0, 0)	(1, -1, 0, 0, 0, 0)	(1, 0, 1, 0, 0, 0)	$\begin{bmatrix} -u_1 & u_0 \end{bmatrix}$
(0, 1, 1, 0, 0, 0)	(1, 0, -1, 0, 0, 0)	(1, 1, 0, 0, 0, 0)	$\begin{bmatrix} -u_1 & u_0 \end{bmatrix}$
(0, 1, 1, 0, 0, 0)	(1, 0, 1, 0, 0, 0)	(1, -1, 0, 0, 0, 0)	$\begin{bmatrix} -u_1 & u_0 \end{bmatrix}$
(0, 1, 1, 0, 0, 0)	(1, 1, 0, 0, 0, 0)	(1, 0, -1, 0, 0, 0)	$\begin{bmatrix} -u_1 & u_0 \end{bmatrix}$
(0, 1, 1, 0, 0, 0)	(2, 0, 0, 0, 0, 0)	(2, 0, 0, 0, 0, 0)	u_0
(0, 2, 0, 0, 0, 0)	(-2, 0, 0, 0, 0, 0)	(-2, 0, 0, 0, 0, 0)	u_0
(0, 2, 0, 0, 0, 0)	(-1, -1, 0, 0, 0, 0)	(-1, 1, 0, 0, 0, 0)	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
(0, 2, 0, 0, 0, 0)	(-1, 0, -1, 0, 0, 0)	(-1, 0, -1, 0, 0, 0)	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
(0, 2, 0, 0, 0, 0)	(-1, 0, 1, 0, 0, 0)	(-1, 0, 1, 0, 0, 0)	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
(0, 2, 0, 0, 0, 0)	(-1, 1, 0, 0, 0, 0)	(-1, -1, 0, 0, 0, 0)	$\begin{bmatrix} -u_0 & -u_1 \end{bmatrix}$
(0, 2, 0, 0, 0, 0)	(0, -2, 0, 0, 0, 0)	(0, 2, 0, 0, 0, 0)	$-u_0$
(0, 2, 0, 0, 0, 0)	(0, -1, -1, 0, 0, 0)	(0, 1, -1, 0, 0, 0)	$\begin{bmatrix} u_1 & -u_0 \end{bmatrix}$
(0, 2, 0, 0, 0, 0)	(0, -1, 1, 0, 0, 0)	(0, 1, 1, 0, 0, 0)	$\begin{bmatrix} -u_1 & -u_0 \end{bmatrix}$
(0, 2, 0, 0, 0, 0)	(0, 0, -2, 0, 0, 0)	(0, 0, -2, 0, 0, 0)	u_0
(0, 2, 0, 0, 0, 0)	(0, 0, 2, 0, 0, 0)	(0, 0, 2, 0, 0, 0)	u_0

Root (α)	Root (β)	$\gamma = \sigma_\alpha(\beta)$	Weyl conjugation coefficient
(0, 2, 0, 0, 0, 0)	(0, 1, -1, 0, 0, 0)	(0, -1, -1, 0, 0, 0)	$u_1 \quad -u_0$
(0, 2, 0, 0, 0, 0)	(0, 1, 1, 0, 0, 0)	(0, -1, 1, 0, 0, 0)	$u_1 \quad u_0$
(0, 2, 0, 0, 0, 0)	(0, 2, 0, 0, 0, 0)	(0, -2, 0, 0, 0, 0)	$-u_0$
(0, 2, 0, 0, 0, 0)	(1, -1, 0, 0, 0, 0)	(1, 1, 0, 0, 0, 0)	$-u_1 \quad u_0$
(0, 2, 0, 0, 0, 0)	(1, 0, -1, 0, 0, 0)	(1, 0, -1, 0, 0, 0)	$u_0 \quad u_1$
(0, 2, 0, 0, 0, 0)	(1, 0, 1, 0, 0, 0)	(1, 0, 1, 0, 0, 0)	$u_0 \quad u_1$
(0, 2, 0, 0, 0, 0)	(1, 1, 0, 0, 0, 0)	(1, -1, 0, 0, 0, 0)	$-u_1 \quad u_0$
(0, 2, 0, 0, 0, 0)	(2, 0, 0, 0, 0, 0)	(2, 0, 0, 0, 0, 0)	u_0
(1, -1, 0, 0, 0, 0)	(-2, 0, 0, 0, 0, 0)	(0, -2, 0, 0, 0, 0)	u_0
(1, -1, 0, 0, 0, 0)	(-1, -1, 0, 0, 0, 0)	(-1, -1, 0, 0, 0, 0)	$u_0 \quad -u_1$
(1, -1, 0, 0, 0, 0)	(-1, 0, -1, 0, 0, 0)	(0, -1, -1, 0, 0, 0)	$u_1 \quad u_0$
(1, -1, 0, 0, 0, 0)	(-1, 0, 1, 0, 0, 0)	(0, -1, 1, 0, 0, 0)	$u_0 \quad u_1$
(1, -1, 0, 0, 0, 0)	(-1, 1, 0, 0, 0, 0)	(1, -1, 0, 0, 0, 0)	$u_1 \quad u_0$
(1, -1, 0, 0, 0, 0)	(0, -2, 0, 0, 0, 0)	(-2, 0, 0, 0, 0, 0)	u_0
(1, -1, 0, 0, 0, 0)	(0, -1, -1, 0, 0, 0)	(-1, 0, -1, 0, 0, 0)	$u_1 \quad u_0$
(1, -1, 0, 0, 0, 0)	(0, -1, 1, 0, 0, 0)	(-1, 0, 1, 0, 0, 0)	$u_0 \quad u_1$
(1, -1, 0, 0, 0, 0)	(0, 0, -2, 0, 0, 0)	(0, 0, -2, 0, 0, 0)	u_0
(1, -1, 0, 0, 0, 0)	(0, 0, 2, 0, 0, 0)	(0, 0, 2, 0, 0, 0)	u_0
(1, -1, 0, 0, 0, 0)	(0, 1, -1, 0, 0, 0)	(1, 0, -1, 0, 0, 0)	$u_1 \quad u_0$
(1, -1, 0, 0, 0, 0)	(0, 1, 1, 0, 0, 0)	(1, 0, 1, 0, 0, 0)	$u_1 \quad u_0$
(1, -1, 0, 0, 0, 0)	(0, 2, 0, 0, 0, 0)	(2, 0, 0, 0, 0, 0)	u_0
(1, -1, 0, 0, 0, 0)	(1, -1, 0, 0, 0, 0)	(-1, 1, 0, 0, 0, 0)	$u_1 \quad u_0$
(1, -1, 0, 0, 0, 0)	(1, 0, -1, 0, 0, 0)	(0, 1, -1, 0, 0, 0)	$u_1 \quad u_0$
(1, -1, 0, 0, 0, 0)	(1, 0, 1, 0, 0, 0)	(0, 1, 1, 0, 0, 0)	$u_1 \quad u_0$
(1, -1, 0, 0, 0, 0)	(1, 1, 0, 0, 0, 0)	(1, 1, 0, 0, 0, 0)	$u_0 \quad -u_1$
(1, -1, 0, 0, 0, 0)	(2, 0, 0, 0, 0, 0)	(0, 2, 0, 0, 0, 0)	u_0
(1, 0, -1, 0, 0, 0)	(-2, 0, 0, 0, 0, 0)	(0, 0, -2, 0, 0, 0)	u_0
(1, 0, -1, 0, 0, 0)	(-1, -1, 0, 0, 0, 0)	(0, -1, -1, 0, 0, 0)	$-u_1 \quad u_0$
(1, 0, -1, 0, 0, 0)	(-1, 0, -1, 0, 0, 0)	(-1, 0, -1, 0, 0, 0)	$u_0 \quad -u_1$
(1, 0, -1, 0, 0, 0)	(-1, 0, 1, 0, 0, 0)	(1, 0, -1, 0, 0, 0)	$u_1 \quad u_0$
(1, 0, -1, 0, 0, 0)	(-1, 1, 0, 0, 0, 0)	(0, 1, -1, 0, 0, 0)	$u_0 \quad u_1$
(1, 0, -1, 0, 0, 0)	(0, -2, 0, 0, 0, 0)	(0, -2, 0, 0, 0, 0)	u_0
(1, 0, -1, 0, 0, 0)	(0, -1, -1, 0, 0, 0)	(-1, -1, 0, 0, 0, 0)	$u_1 \quad -u_0$
(1, 0, -1, 0, 0, 0)	(0, -1, 1, 0, 0, 0)	(1, -1, 0, 0, 0, 0)	$u_1 \quad u_0$
(1, 0, -1, 0, 0, 0)	(0, 0, -2, 0, 0, 0)	(-2, 0, 0, 0, 0, 0)	u_0
(1, 0, -1, 0, 0, 0)	(0, 0, 2, 0, 0, 0)	(2, 0, 0, 0, 0, 0)	u_0
(1, 0, -1, 0, 0, 0)	(0, 1, -1, 0, 0, 0)	(-1, 1, 0, 0, 0, 0)	$u_0 \quad u_1$
(1, 0, -1, 0, 0, 0)	(0, 1, 1, 0, 0, 0)	(1, 1, 0, 0, 0, 0)	$u_1 \quad -u_0$
(1, 0, -1, 0, 0, 0)	(0, 2, 0, 0, 0, 0)	(0, 2, 0, 0, 0, 0)	u_0
(1, 0, -1, 0, 0, 0)	(1, -1, 0, 0, 0, 0)	(0, -1, 1, 0, 0, 0)	$u_1 \quad u_0$
(1, 0, -1, 0, 0, 0)	(1, 0, -1, 0, 0, 0)	(-1, 0, 1, 0, 0, 0)	$u_1 \quad u_0$
(1, 0, -1, 0, 0, 0)	(1, 0, 1, 0, 0, 0)	(1, 0, 1, 0, 0, 0)	$u_0 \quad -u_1$
(1, 0, -1, 0, 0, 0)	(1, 1, 0, 0, 0, 0)	(0, 1, 1, 0, 0, 0)	$-u_1 \quad u_0$
(1, 0, -1, 0, 0, 0)	(2, 0, 0, 0, 0, 0)	(0, 0, 2, 0, 0, 0)	u_0
(1, 0, 1, 0, 0, 0)	(-2, 0, 0, 0, 0, 0)	(0, 0, 2, 0, 0, 0)	$-u_0$
(1, 0, 1, 0, 0, 0)	(-1, -1, 0, 0, 0, 0)	(0, -1, 1, 0, 0, 0)	$u_0 \quad -u_1$
(1, 0, 1, 0, 0, 0)	(-1, 0, -1, 0, 0, 0)	(1, 0, 1, 0, 0, 0)	$-u_0 \quad u_1$
(1, 0, 1, 0, 0, 0)	(-1, 0, 1, 0, 0, 0)	(-1, 0, 1, 0, 0, 0)	$-u_0 \quad u_1$
(1, 0, 1, 0, 0, 0)	(-1, 1, 0, 0, 0, 0)	(0, 1, 1, 0, 0, 0)	$u_1 \quad -u_0$
(1, 0, 1, 0, 0, 0)	(0, -2, 0, 0, 0, 0)	(0, -2, 0, 0, 0, 0)	u_0

Root (α)	Root (β)	$\gamma = \sigma_\alpha(\beta)$	Weyl conjugation coefficient
(1, 0, 1, 0, 0, 0)	(0, -1, -1, 0, 0, 0)	(1, -1, 0, 0, 0, 0)	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
(1, 0, 1, 0, 0, 0)	(0, -1, 1, 0, 0, 0)	(-1, -1, 0, 0, 0, 0)	$\begin{bmatrix} -u_0 & u_1 \end{bmatrix}$
(1, 0, 1, 0, 0, 0)	(0, 0, -2, 0, 0, 0)	(2, 0, 0, 0, 0, 0)	$-u_0$
(1, 0, 1, 0, 0, 0)	(0, 0, 2, 0, 0, 0)	(-2, 0, 0, 0, 0, 0)	$-u_0$
(1, 0, 1, 0, 0, 0)	(0, 1, -1, 0, 0, 0)	(1, 1, 0, 0, 0, 0)	$\begin{bmatrix} -u_0 & -u_1 \end{bmatrix}$
(1, 0, 1, 0, 0, 0)	(0, 1, 1, 0, 0, 0)	(-1, 1, 0, 0, 0, 0)	$\begin{bmatrix} u_1 & -u_0 \end{bmatrix}$
(1, 0, 1, 0, 0, 0)	(0, 2, 0, 0, 0, 0)	(0, 2, 0, 0, 0, 0)	u_0
(1, 0, 1, 0, 0, 0)	(1, -1, 0, 0, 0, 0)	(0, -1, -1, 0, 0, 0)	$\begin{bmatrix} -u_0 & -u_1 \end{bmatrix}$
(1, 0, 1, 0, 0, 0)	(1, 0, -1, 0, 0, 0)	(1, 0, -1, 0, 0, 0)	$\begin{bmatrix} u_0 & -u_1 \end{bmatrix}$
(1, 0, 1, 0, 0, 0)	(1, 0, 1, 0, 0, 0)	(-1, 0, -1, 0, 0, 0)	$\begin{bmatrix} -u_0 & u_1 \end{bmatrix}$
(1, 0, 1, 0, 0, 0)	(1, 1, 0, 0, 0, 0)	(0, 1, -1, 0, 0, 0)	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
(1, 0, 1, 0, 0, 0)	(2, 0, 0, 0, 0, 0)	(0, 0, -2, 0, 0, 0)	$-u_0$
(1, 1, 0, 0, 0, 0)	(-2, 0, 0, 0, 0, 0)	(0, 2, 0, 0, 0, 0)	$-u_0$
(1, 1, 0, 0, 0, 0)	(-1, -1, 0, 0, 0, 0)	(1, 1, 0, 0, 0, 0)	$\begin{bmatrix} -u_0 & u_1 \end{bmatrix}$
(1, 1, 0, 0, 0, 0)	(-1, 0, -1, 0, 0, 0)	(0, 1, -1, 0, 0, 0)	$\begin{bmatrix} u_0 & -u_1 \end{bmatrix}$
(1, 1, 0, 0, 0, 0)	(-1, 0, 1, 0, 0, 0)	(0, 1, 1, 0, 0, 0)	$\begin{bmatrix} -u_1 & -u_0 \end{bmatrix}$
(1, 1, 0, 0, 0, 0)	(-1, 1, 0, 0, 0, 0)	(-1, 1, 0, 0, 0, 0)	$\begin{bmatrix} -u_0 & u_1 \end{bmatrix}$
(1, 1, 0, 0, 0, 0)	(0, -2, 0, 0, 0, 0)	(2, 0, 0, 0, 0, 0)	$-u_0$
(1, 1, 0, 0, 0, 0)	(0, -1, -1, 0, 0, 0)	(1, 0, -1, 0, 0, 0)	$\begin{bmatrix} -u_0 & u_1 \end{bmatrix}$
(1, 1, 0, 0, 0, 0)	(0, -1, 1, 0, 0, 0)	(1, 0, 1, 0, 0, 0)	$\begin{bmatrix} -u_0 & -u_1 \end{bmatrix}$
(1, 1, 0, 0, 0, 0)	(0, 0, -2, 0, 0, 0)	(0, 0, -2, 0, 0, 0)	u_0
(1, 1, 0, 0, 0, 0)	(0, 0, 2, 0, 0, 0)	(0, 0, 2, 0, 0, 0)	u_0
(1, 1, 0, 0, 0, 0)	(0, 1, -1, 0, 0, 0)	(-1, 0, -1, 0, 0, 0)	$\begin{bmatrix} -u_0 & u_1 \end{bmatrix}$
(1, 1, 0, 0, 0, 0)	(0, 1, 1, 0, 0, 0)	(-1, 0, 1, 0, 0, 0)	$\begin{bmatrix} u_1 & u_0 \end{bmatrix}$
(1, 1, 0, 0, 0, 0)	(0, 2, 0, 0, 0, 0)	(-2, 0, 0, 0, 0, 0)	$-u_0$
(1, 1, 0, 0, 0, 0)	(1, -1, 0, 0, 0, 0)	(1, -1, 0, 0, 0, 0)	$\begin{bmatrix} u_0 & -u_1 \end{bmatrix}$
(1, 1, 0, 0, 0, 0)	(1, 0, -1, 0, 0, 0)	(0, -1, -1, 0, 0, 0)	$\begin{bmatrix} u_0 & -u_1 \end{bmatrix}$
(1, 1, 0, 0, 0, 0)	(1, 0, 1, 0, 0, 0)	(0, -1, 1, 0, 0, 0)	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
(1, 1, 0, 0, 0, 0)	(1, 1, 0, 0, 0, 0)	(-1, -1, 0, 0, 0, 0)	$\begin{bmatrix} -u_0 & u_1 \end{bmatrix}$
(1, 1, 0, 0, 0, 0)	(2, 0, 0, 0, 0, 0)	(0, -2, 0, 0, 0, 0)	$-u_0$
(2, 0, 0, 0, 0, 0)	(-2, 0, 0, 0, 0, 0)	(2, 0, 0, 0, 0, 0)	$-u_0$
(2, 0, 0, 0, 0, 0)	(-1, -1, 0, 0, 0, 0)	(1, -1, 0, 0, 0, 0)	$\begin{bmatrix} -u_1 & u_0 \end{bmatrix}$
(2, 0, 0, 0, 0, 0)	(-1, 0, -1, 0, 0, 0)	(1, 0, -1, 0, 0, 0)	$\begin{bmatrix} -u_1 & u_0 \end{bmatrix}$
(2, 0, 0, 0, 0, 0)	(-1, 0, 1, 0, 0, 0)	(1, 0, 1, 0, 0, 0)	$\begin{bmatrix} -u_0 & -u_1 \end{bmatrix}$
(2, 0, 0, 0, 0, 0)	(-1, 1, 0, 0, 0, 0)	(1, 1, 0, 0, 0, 0)	$\begin{bmatrix} -u_0 & -u_1 \end{bmatrix}$
(2, 0, 0, 0, 0, 0)	(0, -2, 0, 0, 0, 0)	(0, -2, 0, 0, 0, 0)	u_0
(2, 0, 0, 0, 0, 0)	(0, -1, -1, 0, 0, 0)	(0, -1, -1, 0, 0, 0)	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
(2, 0, 0, 0, 0, 0)	(0, -1, 1, 0, 0, 0)	(0, -1, 1, 0, 0, 0)	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
(2, 0, 0, 0, 0, 0)	(0, 0, -2, 0, 0, 0)	(0, 0, -2, 0, 0, 0)	u_0
(2, 0, 0, 0, 0, 0)	(0, 0, 2, 0, 0, 0)	(0, 0, 2, 0, 0, 0)	u_0
(2, 0, 0, 0, 0, 0)	(0, 1, -1, 0, 0, 0)	(0, 1, -1, 0, 0, 0)	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
(2, 0, 0, 0, 0, 0)	(0, 1, 1, 0, 0, 0)	(0, 1, 1, 0, 0, 0)	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
(2, 0, 0, 0, 0, 0)	(0, 2, 0, 0, 0, 0)	(0, 2, 0, 0, 0, 0)	u_0
(2, 0, 0, 0, 0, 0)	(1, -1, 0, 0, 0, 0)	(-1, -1, 0, 0, 0, 0)	$\begin{bmatrix} -u_1 & u_0 \end{bmatrix}$
(2, 0, 0, 0, 0, 0)	(1, 0, -1, 0, 0, 0)	(-1, 0, -1, 0, 0, 0)	$\begin{bmatrix} -u_1 & u_0 \end{bmatrix}$
(2, 0, 0, 0, 0, 0)	(1, 0, 1, 0, 0, 0)	(-1, 0, 1, 0, 0, 0)	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
(2, 0, 0, 0, 0, 0)	(1, 1, 0, 0, 0, 0)	(-1, 1, 0, 0, 0, 0)	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
(2, 0, 0, 0, 0, 0)	(2, 0, 0, 0, 0, 0)	(-2, 0, 0, 0, 0, 0)	$-u_0$

6.3 Weyl group conjugation identities

Below is an explicit formulation of the Weyl group conjugation identities.

WeylConjugationIdentitiesPlaceholder